



AKTU B.Tech I-Year



FME

ONE SHOT Revision

(Crash Course)

Unit- 4

Introduction to Fluids Mechanics and its applications

Pdf Notes + AKTU PYQs+ Imp Qns

By M S Tomar Sir



Introduction

✓
"Matter" is anything that has mass and occupies space.

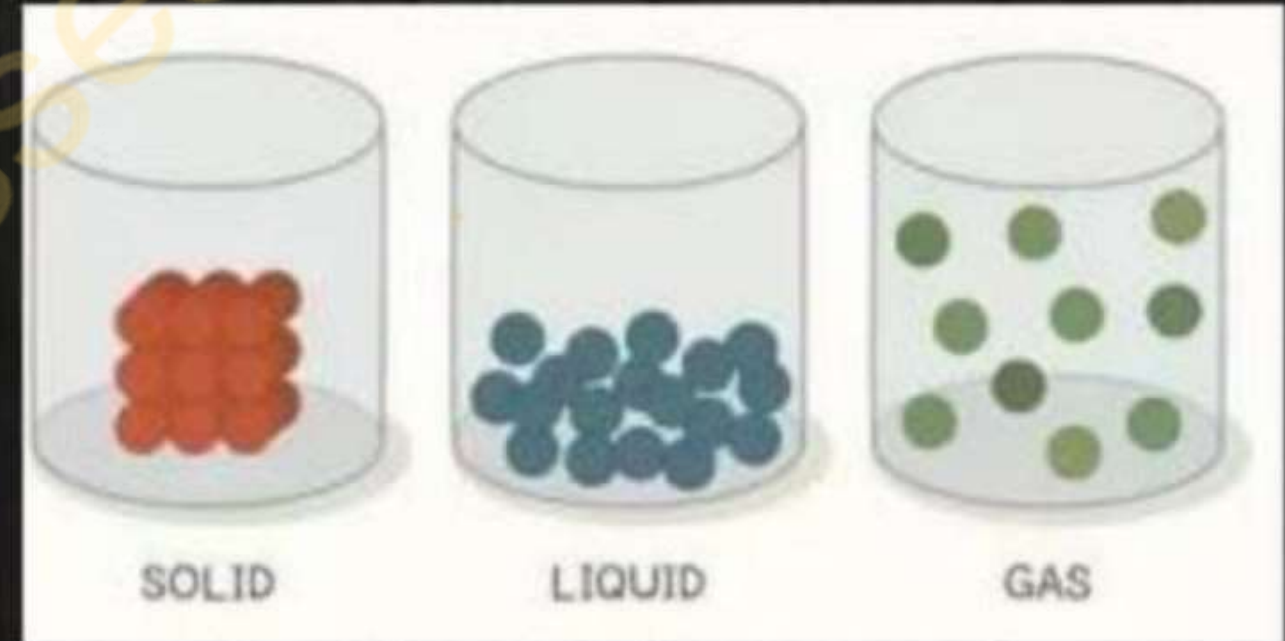
A matter exists in either the solid state or the fluid state.

The fluid state is further divided into

- Liquid state
- Gaseous state
- In fact the same matter may exist in any one of the three states i.e. solid, liquid and gaseous.
- For example water occurs in a liquid state, may also occur in solid state as ice and in a gaseous state as vapour.

.....introduction

- In solids, the molecules are very closely spaced.
- In liquids, the spacing between the molecules is relatively large.
- In gases, the space between the molecules is still larger.



In general

$$\rho_{\text{solids}} > \rho_{\text{liquids}} > \rho_{\text{gases}}$$

Fluid

- Fluid may be defined as a substance which is capable of flowing.
- It has no definite shape of its own but conforms to the shape of the containing vessel.

OR

- “A fluid is a substance which deforms continuously under the action of tangential or shear force.”
- Examples of fluids are Water, Milk, Kerosene, Petrol, Gases, etc.

Fluid Mechanics

“Fluid mechanics is that branch of science which deals with the behavior of the fluids at rest as well as in motion.”



- In general the scope of fluid mechanics is very wide which includes the study of all liquids and gases.
- It has applications in mechanical, civil, chemical and biomedical engineering etc.

Basic Properties of Fluid

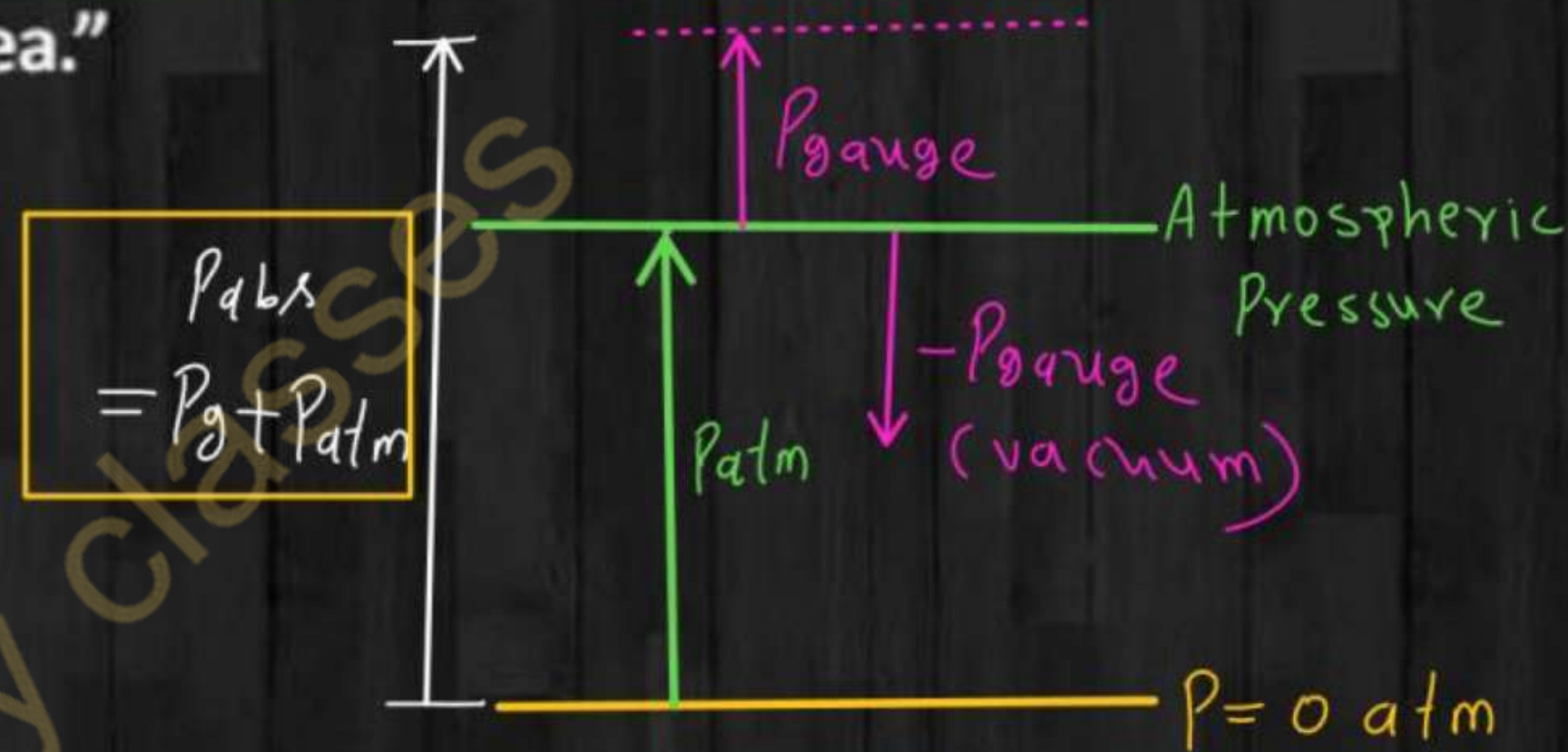
1. Pressure
2. Density
3. Specific Weight
4. Specific Gravity
5. Dynamic Viscosity
6. Kinematic Viscosity

1. Pressure

"It is defined as normal force per unit area."

$$P = \frac{F}{A}$$

- It is a scalar quantity.
- Units of Pressure :
- N/m^2
- Pascal (Pa) $(1 \text{ Pa} = 1 \text{ N/m}^2)$
- atm $(1 \text{ atm} = 101325 \text{ Pa} = 101.325 \text{ kPa})$
- Bar $(1 \text{ bar} = 10^5 \text{ Pa} = 10^5 \text{ N/m}^2)$



Numerical-1 What is the absolute pressure experienced by a pressure sensor, if the atmospheric pressure of a fluid is 2 atm, gauge pressure is 5 atm and differential pressure is 3 atm.

AKTU : 2021-22

given:

$$P_{atm} = 2 \text{ atm}$$

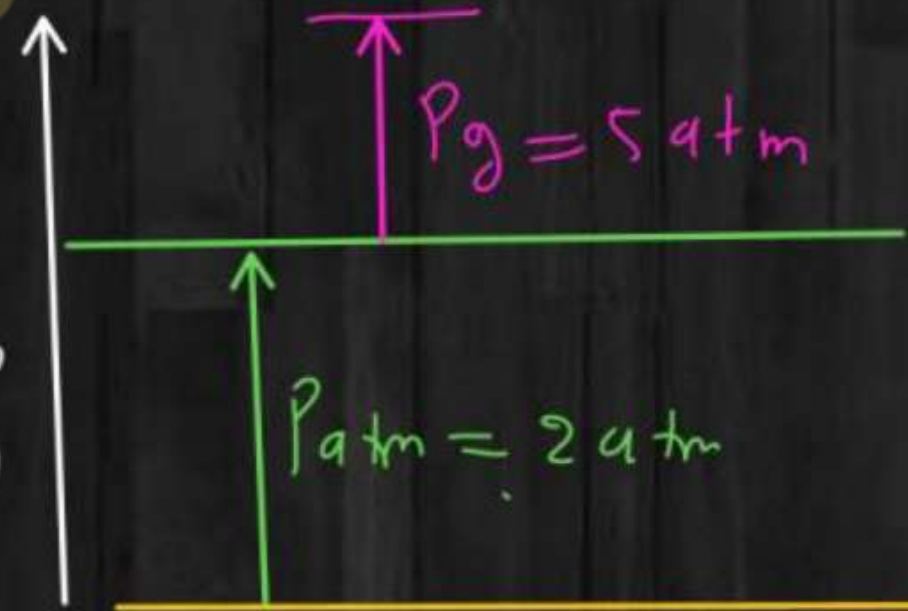
$$P_g = 5 \text{ atm}$$

$$P_{abs} = ?$$

$$\begin{aligned} P_{abs} &= P_g + P_{atm} \\ &= 5 + 2 \end{aligned}$$

$$P_{abs} = 7 \text{ atm} \quad \text{Ans}$$

$$\begin{aligned} P_{abs} &= P_{atm} \\ &+ P_g \end{aligned}$$



2. Density or Mass Density (ρ)

"It is defined as mass per unit volume of the substance."

$$\rho = \frac{m}{V}$$

Ex.

$$\rho = \frac{m}{V} = \frac{100 \text{ kg}}{75 \text{ m}^3}$$

$$\rho = 1.33 \text{ kg/m}^3$$

Room air

$$m = 100 \text{ kg}$$

$$V = 75 \text{ m}^3$$

$$\rho = ?$$

Units of density

- Kg/m³
- g/cm³ or g/cc (1 g/cc = 10³ kg/m³)

For example

- $\rho_{\text{steel}} = 7850 \text{ kg/m}^3$
- $\rho_{\text{water}} = 1000 \text{ kg/m}^3$
- $\rho_{\text{air}} = 1.2 \text{ kg/m}^3$

$$\rho_s > \rho_L > \rho_a$$

specific volume (v)

$$v = \frac{V}{m} = \frac{1}{\left(\frac{m}{V}\right)} = \frac{1}{\rho}$$

$$v = \frac{1}{\rho}$$

$$= \frac{75}{100} \text{ m}^3/\text{kg}$$

3. Specific Weight or weight density (ω)

“It is defined as weight per unit volume of substance.”

$$\omega = \frac{\text{weight}}{\text{Volume}} = \frac{mg}{V} = \rho g$$

$$\left[\rho = \frac{m}{V} \right]$$

$$g = 9.81 \text{ m/s}^2$$

$$\text{sp. wt. } (\omega) = \rho g$$

➤ Unit of specific weight is N/m^3

4. Specific Gravity(s)

“ It is defined as the density of the fluid w.r.t. the density of a standard fluid.”

$$\text{Specific Gravity(s)} = \frac{\text{density of fluid}}{\text{density of standard fluid}} = \frac{\rho_{\text{fluid}}}{\rho_{\text{std fluid}}}$$

- For liquid, standard fluid is water at 4°C(39.2°F), 1 atm, ($\rho = 1000 \text{ kg/m}^3$).
- For gases, standard fluid is Air.

Numerical-2 Calculate the specific weight, density and specific gravity of one litre of a liquid which weighs 7 N.

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Solution

given

Weight = 7 N

Volume = 1 litre

$$V = 1 \times 10^{-3} \text{ m}^3$$

Determine (i) sp. weight (w) = ?

(ii) density (ρ) = ?

(iii) sp. gravity (S) = ?

$$(i) \text{ sp. wt} = \frac{W}{V} = \frac{7 \text{ N}}{1 \times 10^{-3} \text{ m}^3}$$

$$\text{sp. wt} = 7000 \text{ N/m}^3 \text{ Ans}$$

$$(ii) \text{ sp. wt} = \rho \cdot g \Rightarrow 7000 = \rho \times 9.81$$

$$\Rightarrow \rho = 713.56 \text{ kg/m}^3 \text{ Ans}$$

$$(iii) \text{ sp. gravity} = \frac{\rho_{\text{fluid}}}{\rho_{\text{std. fluid}}} = \frac{713.56 \text{ kg/m}^3}{1000 \text{ kg/m}^3}$$

$$\text{sp. gr.} = 0.713 \text{ Ans}$$

Numerical-3 Calculate the density, specific weight of one litre of petrol of specific gravity = 0.7.

Try yourself

Ans density, $\rho = 700 \text{ kg/m}^3$
specific weight, $w = 6867 \frac{\text{N}}{\text{m}^3}$

Gateway classes

5. Dynamic Viscosity

"Two adjacent layers of the fluid resist the motion of each other such a fundamental property of the fluid is known as viscosity or dynamic viscosity."

➤ Therefore the frictional force between the adjacent layers is known as viscous shear force.

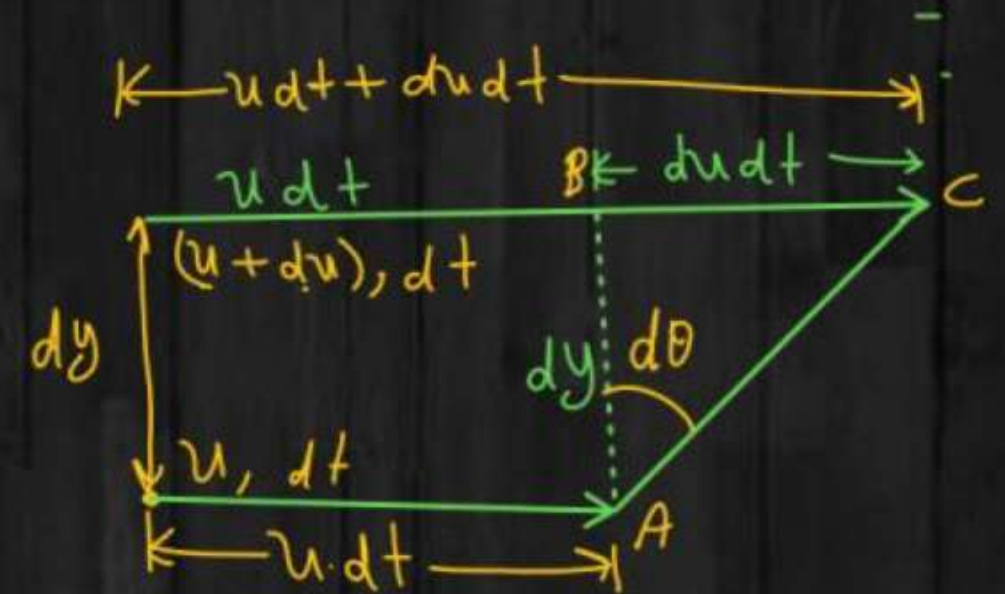
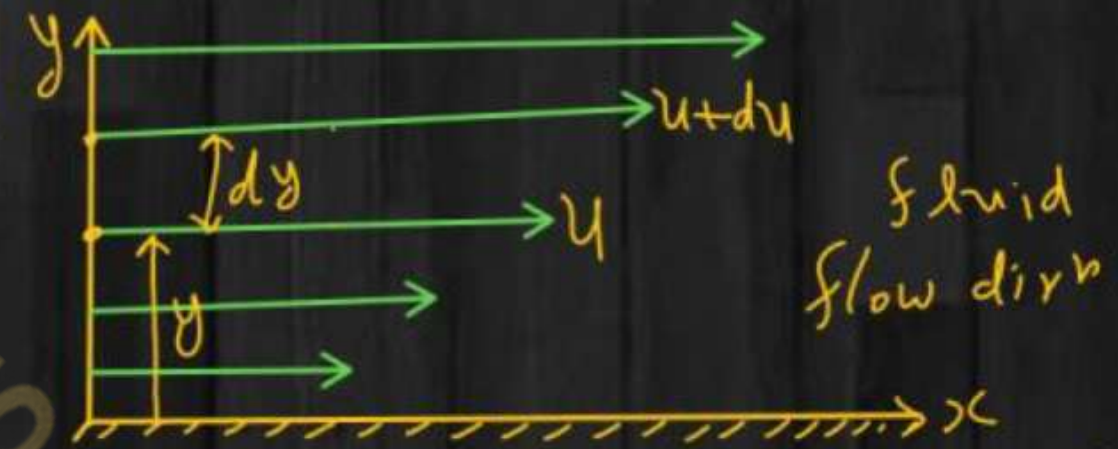
$$\tan \Delta\theta \approx \Delta\theta = \frac{du}{dy} dt$$

$$\because \Delta\theta \downarrow \downarrow \downarrow \Rightarrow \tan \Delta\theta \approx \Delta\theta$$

$$\Delta\theta = \frac{du}{dy} dt \Rightarrow \boxed{\frac{d\theta}{dt} = \frac{du}{dy}}$$

Where, $\frac{d\theta}{dt}$ = Rate of angular (shear) deformation and

$\frac{du}{dy}$ = Velocity gradient.



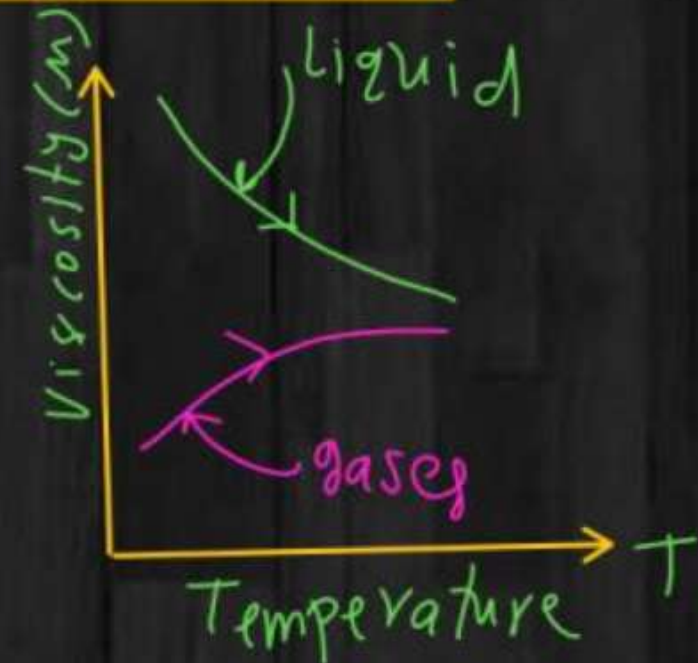
Cause of Viscosity

- Basic cause of viscosity is **cohesive forces** between the molecules.
- Because in case of liquid cohesive force is high and in case of gases cohesive force is very less.
- Therefore viscosity of liquid is very high as compared to viscosity of gases.

Dependency of Viscosity on Temperature

2 marks

- The viscosity of **liquid** decreases with increase in temperature. [if $T \uparrow$ then cohesion $\downarrow$]
- The viscosity of **gases** increases with increase in temperature. [if $T \uparrow$ then randomness $\uparrow$]



6. Kinematic Viscosity

“Ratio of dynamic Viscosity and density is called Kinematic Viscosity.

$$\nu = \frac{\mu}{\rho} = \frac{\text{dynamic viscosity}}{\text{density}}$$

Unit of Kinematic Viscosity

➤ MKS : m^2/s

➤ CGS : cm^2/s

➤ $1 \frac{\text{cm}^2}{\text{s}} = 10^{-4} \frac{\text{m}^2}{\text{s}} = 1 \text{ stoke}$

$$\begin{aligned} \nu &= \frac{\mu}{\rho} = \frac{\frac{\text{N} \cdot \text{s}}{\text{m}^2}}{\frac{\text{kg}}{\text{m}^3}} \\ &= \frac{\frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \frac{\text{s}}{\text{m}^2}}{\text{kg}}}{\frac{\text{kg}}{\text{m}^3}} = \frac{\text{m}^2}{\text{s}} \end{aligned}$$

$$\begin{aligned} \text{N} &\Rightarrow F = ma \\ &= \text{kg} \cdot \frac{\text{m}}{\text{s}^2} \end{aligned}$$

Newton's law of Viscosity

According to Newton's law of viscosity "Shear stress between the layers of fluid is directly proportional to rate of shear deformation."

$$\tau \propto \frac{d\theta}{dt} \Rightarrow \tau \propto \frac{du}{dy}$$

$$\tau = \mu \frac{du}{dy}$$

$\Rightarrow$

$$\frac{F}{A} = \mu \frac{du}{dy}$$



Where μ is the proportionality constant (property of fluid) which is known as viscosity or dynamic viscosity.

Unit of Viscosity

$$\tau \propto \frac{d\theta}{dt} \Rightarrow \tau \propto \frac{du}{dy}$$

$$\tau = \mu \cdot \frac{du}{dy}$$

$$N \Rightarrow \text{Force} \Rightarrow F = ma$$

$$= \text{kg} \cdot \frac{\text{m}}{\text{s}^2}$$

$$\mu = \frac{\tau}{\frac{du}{dy}} = \frac{\frac{N}{m^2}}{\frac{\text{m/s}}{\text{m}}} = \frac{N \cdot s}{m^2} = \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{s}}{\text{m}^2} = \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

Unit of viscosity (μ)

➤ MKS: $\frac{N \cdot s}{m^2} = \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{s}}{\text{m}^2} = \frac{\text{kg}}{\text{m} \cdot \text{s}}$ [F = ma]

➤ CGS: $\frac{\text{gm}}{\text{cm} \cdot \text{s}} = \frac{10^{-3} \text{kg}}{10^{-2} \text{m} \cdot \text{s}} = 0.1 \frac{\text{kg}}{\text{m} \cdot \text{s}} = 0.1 \frac{N \cdot s}{m^2}$

❖ $\frac{\text{gm}}{\text{cm} \cdot \text{s}} = 1 \text{ Poise} = 0.1 \frac{N \cdot s}{m^2}$

Types of Fluid

The fluid may be classified into the following five types:

- Newtonian fluid ✓
- Non-Newtonian fluid ✓
- Ideal fluid ✓
- Real fluid ✓
- Ideal plastic fluid

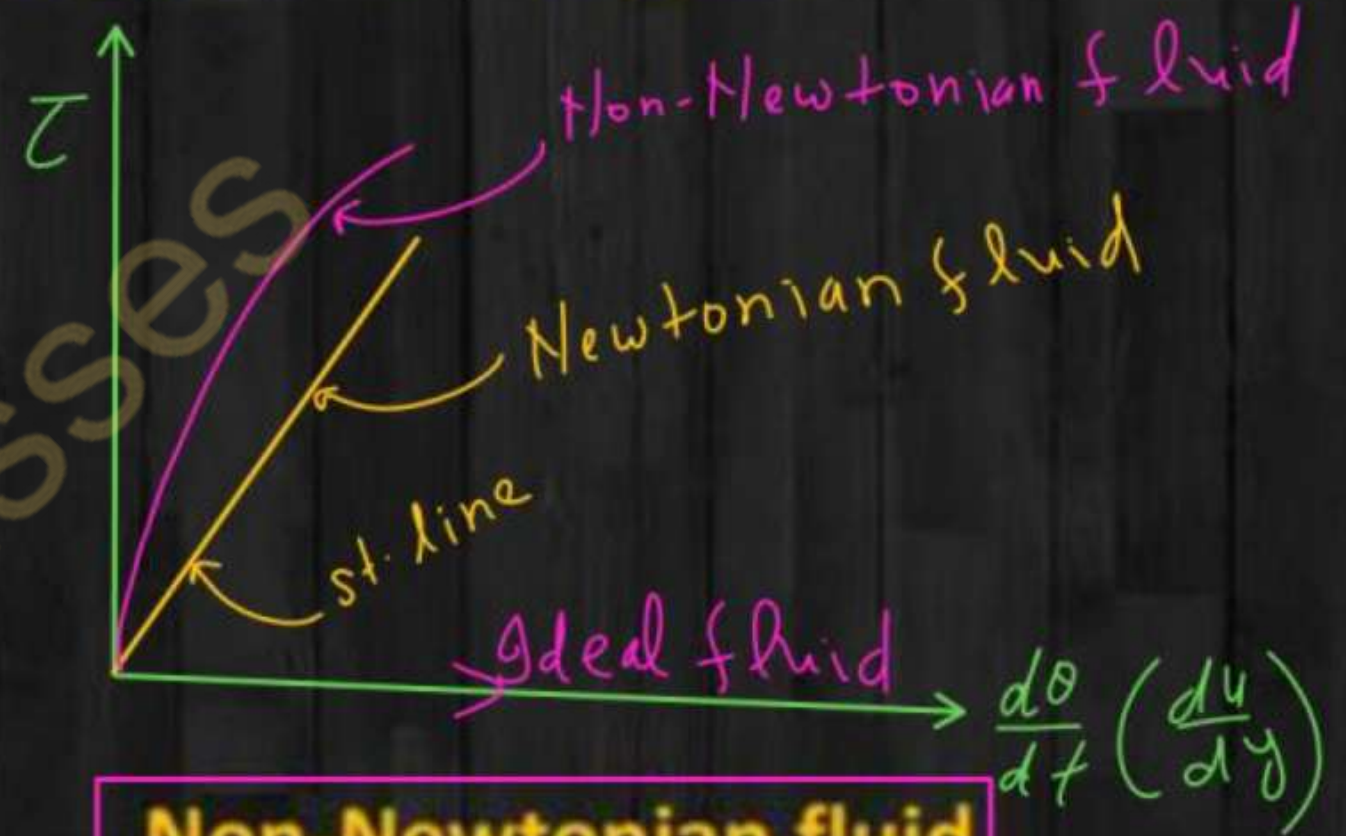
Newtonian fluid Non-Newtonian fluid

Newtonian fluid

- All the fluids which obey Newton's law of viscosity are known as Newtonian fluids.
- There is a linear relation between magnitude of τ and $\frac{d\theta}{dt}$. $(\tau \propto \frac{d\theta}{dt})$

Example : Air, water, kerosene, petrol etc.

- $\tau = \mu \frac{du}{dy}$ Or $\tau = \mu \frac{d\theta}{dt}$
- $\mu = \tau / (\frac{d\theta}{dt})$



Non-Newtonian fluid

The fluids which do not follow Newton's law viscosity are known as Non-Newtonian Fluid.

Example : Blood, paint, mud etc.

There is a non-linear relation between magnitude of τ and $\frac{d\theta}{dt}$.

IDEAL FLUID and REAL FLUID

IDEAL FLUID:- Incompressible fluid having zero viscosity is called ideal fluid ($\tau = 0$).

- Ideal fluids do not exist in nature.

REAL FLUID: A fluid, which possess viscosity is known as real fluid. All the fluids, in actual practice, are real fluids.

Examples : Air, water, kerosene, petrol etc.

$$\tau = \mu \frac{du}{dy} \quad [\mu = 0]$$

$$\tau = 0$$

$$\mu > 0, \tau > 0$$

Numerical-1 How does temperature affect the viscosity of a fluid? A square plate 0.1 m side moves parallel to second plate with a velocity of 0.1 ms^{-1} , both plates being immersed in water. If the viscous force is 0.002 N and the coefficient of viscosity 0.001 poise , what is the distance between the plates ?

AKTU : 2023-24 Odd

Numerical

given:

$$A = 0.1 \times 0.1 \text{ m}^2$$

$$A = 0.01 \text{ m}^2$$

$$du = 0.1 \text{ m/s}$$

$$F = 0.002 \text{ N}$$

$$\mu = 0.001 \text{ poise}$$

$$= 0.001 \times 0.1$$

$$= 10^{-4} \text{ N.s/m}^2$$

$$dy = ?$$

$$\tau = \mu \frac{du}{dy}$$

$$\frac{F}{A} = \mu \frac{du}{dy}$$

$$\frac{0.002}{0.01} = 10^{-4} \times \frac{0.1}{dy}$$

$$dy = 5 \times 10^{-5} \text{ m}$$

$$dy = 0.05 \text{ mm}$$

Numerical-2 A metal plate of area $2.5 \times 10^{-4} \text{ m}^2$ is placed on a $0.25 \times 10^{-3} \text{ m}$ thick layer of castor oil. If a force of 2.5 N is needed to move the plate with a velocity $3 \times 10^{-2} \text{ ms}^{-1}$, calculate the coefficient of viscosity of castor oil.

given:

$$A = 2.5 \times 10^{-4} \text{ m}^2$$

$$dy = 0.25 \times 10^{-3} \text{ m}$$

$$F = 2.5 \text{ N}$$

$$du = 3 \times 10^{-2} \text{ m/s}$$

$$\mu = ?$$

$$\tau = \mu \frac{du}{dy}$$

$$\frac{F}{A} = \mu \frac{du}{dy}$$

$$\frac{2.5}{2.5 \times 10^{-4}} = \mu \times \frac{3 \times 10^{-2}}{0.25 \times 10^{-3}}$$

$$\mu = 83.33 \frac{\text{N}\cdot\text{s}}{\text{m}^2}$$

Ans

Numerical-3 The shear stress at a point in a liquid is found to be 0.03 N/m^2 . The velocity gradient at the point is 0.21 s^{-1} . What will be its viscosity?

given

$$\tau = 0.03 \text{ N/m}^2$$

$$\frac{du}{dy} = 0.21 \text{ s}^{-1}$$

$$\mu = ?$$

$$\tau = \mu \frac{du}{dy}$$

$$0.03 = \mu \times 0.21$$

$$\mu = 0.1428 \frac{\text{N} \cdot \text{s}}{\text{m}^2}$$

Ans

Numerical-4 Water is flowing slowly on a horizontal plane, the viscosity coefficient of water is 0.01 poise, and its surface area is 100 cm^2 . What is the external force required to maintain the velocity gradient of the flow 1 s^{-1} ?

Ans. 10^{-5} N

Try Yourself

Gateway classes

Numerical-5 Calculate the horizontal force required to move a metal plate of area $2 \times 10^{-2} \text{ m}^2$ with a velocity of $4.5 \times 10^{-2} \text{ m s}^{-1}$ when it rests on a layer of oil $1.5 \times 10^{-3} \text{ m}$ thick. $\eta = 2 \text{ Nsm}^{-2}$.

Ans. 1.2 N

Try Yourself

Gateway classes

Numerical-6 A flat plate is separated from a large plate by a layer of glycerine of thickness 3×10^{-3} m. If the coefficient of viscosity of glycerine 2 Ns/m^2 what is the force required to keep the plate moving with a velocity of $6 \times 10^{-2} \text{ m/s}$. Area of the plate is $4.8 \times 10^{-3} \text{ m}^2$.

Try Yourself

Ans. 0.192 N

Gateway classes

Pascal's Law

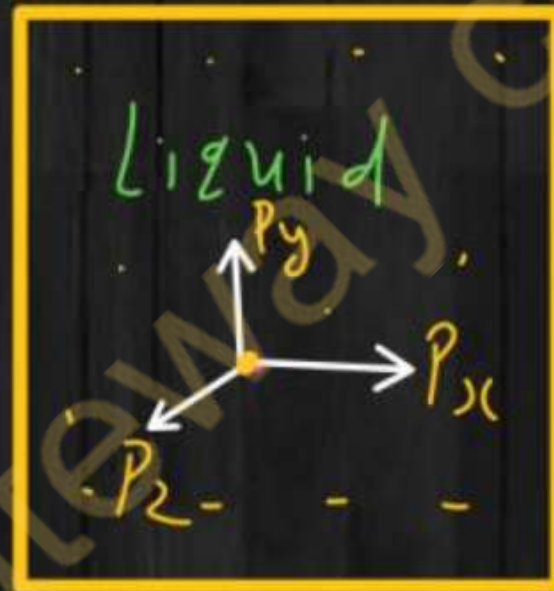
Statement: Pascal's Law states that in a closed system, the pressure applied to a confined fluid is transmitted equally in all directions throughout the fluid.

"Pressure at a point in fluid at rest is same in all the directions".

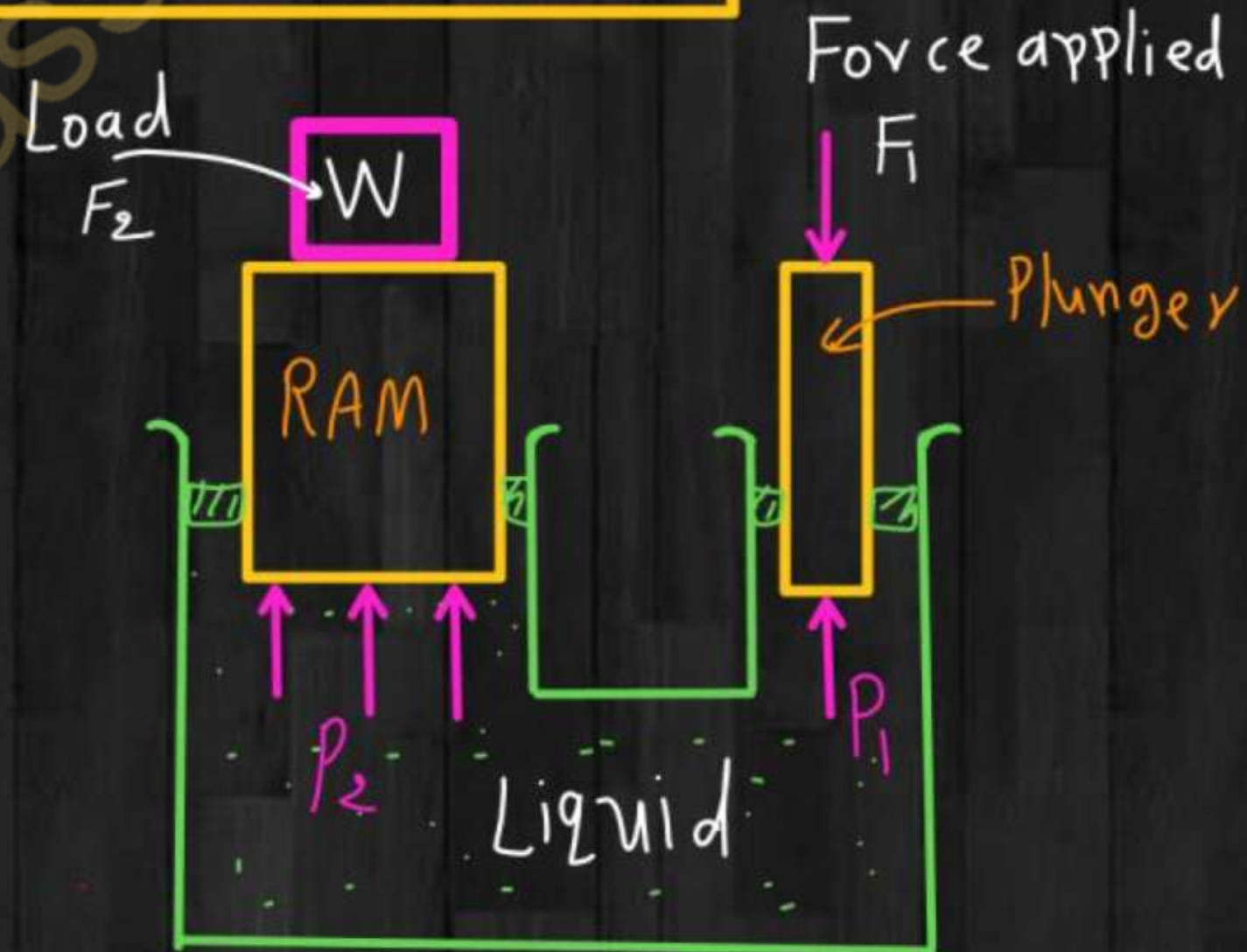
$$P = \frac{F}{A}, P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Applications of Pascal's law

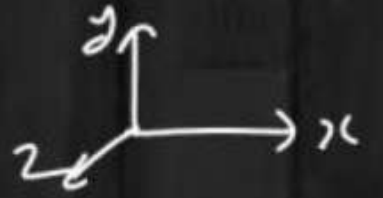
- Hydraulic lift ✓
- Hydraulic Jacks ✓
- Hydraulic brakes ✓
- Hydraulic pumps ✓



$$P_x = P_y = P_z$$



Pascal's Law Derivation



Let us consider a wedge shape fluid element whose thickness is unity(1).

$$\sum F_x = 0 \Rightarrow \rightarrow (+) \quad \leftarrow (-)$$

$$P_x \cdot dy \cdot 1 - P_z \cdot ds \cdot 1 \cdot \cos\theta = 0$$

$$P_x \cdot dy = P_z \cdot ds \cdot \cos\theta$$

$$P_x \cdot dy = P_z \cdot dx$$

$$P_x = P_z \quad \dots (i)$$

$$\sum F_y = 0 \quad \uparrow (+) \quad \downarrow (-)$$

$$P_y \cdot dx \cdot 1 - P_z \cdot ds \cdot 1 \cdot \sin\theta - mg$$

$$P_y \cdot dx = P_z \cdot ds \cdot \sin\theta + \frac{1}{2} dx \cdot dy \cdot \rho \cdot g$$

$$P_y \cdot dx = P_z \cdot dx$$

$$\Rightarrow P_y = P_z \quad \dots (ii)$$

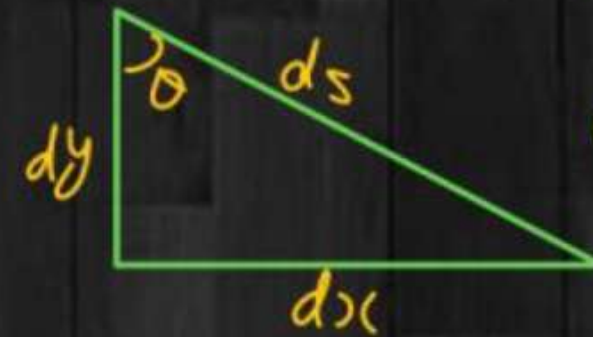
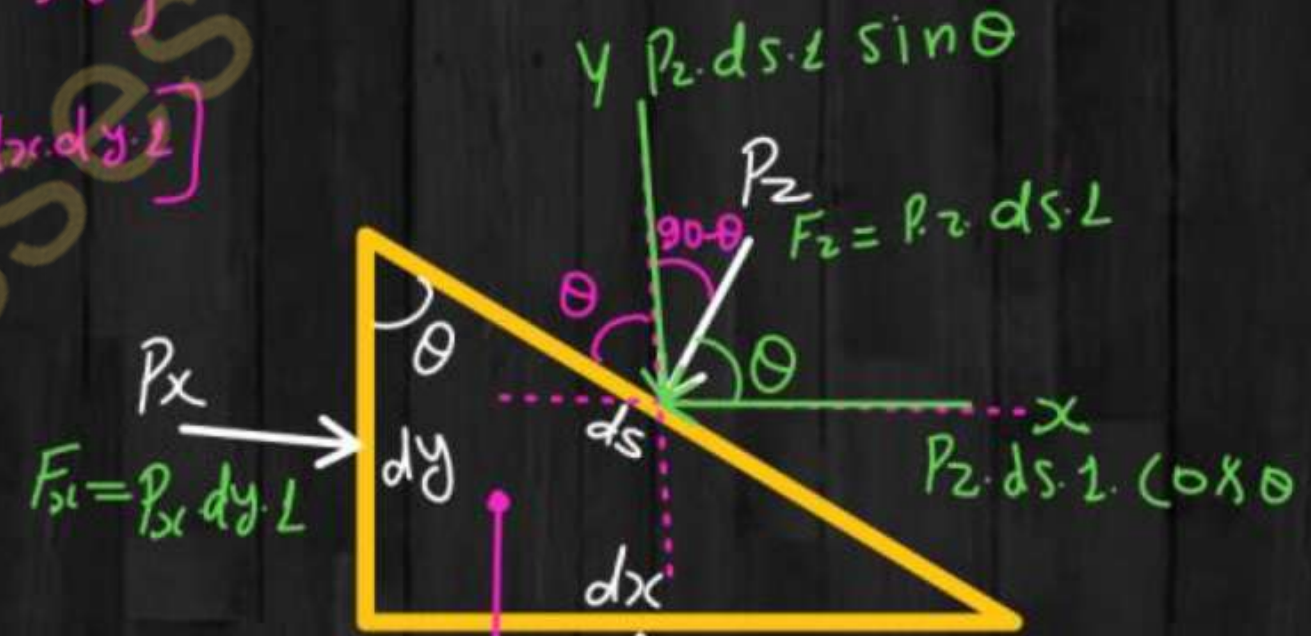
$$W = mg \quad \left[\rho = \frac{m}{V} \Rightarrow m = \rho \cdot V \right]$$

$$W = \rho \cdot V \cdot g \quad \left[V = A \cdot L = \frac{1}{2} dx \cdot dy \cdot 1 \right]$$

$$W = \frac{1}{2} dx \cdot dy \cdot \rho \cdot g = mg$$

From eqn (i) & (ii)

$$P_x = P_y = P_z$$



$$\cos\theta = \frac{dy}{ds} \Rightarrow dy = ds \cos\theta$$

$$\sin\theta = \frac{dx}{ds} \Rightarrow dx = ds \sin\theta$$

Pascal's Law

Numerical.1 Explain Pascal's law with help of a neat sketch. What are its practical applications? The small piston of a hydraulic lift has an area of 0.20 m^2 . A car weighing $1.2 \times 10^4 \text{ N}$ sits on a rack mounted on the large piston. The large piston has an area of 0.90 m^2 . How large force must be applied to the small piston to support the car.

AKTU : 2022-23 Odd

given:

$$A_1 = 0.2 \text{ m}^2$$

$$F_2 = W = 1.2 \times 10^4 \text{ N}$$

$$A_2 = 0.9 \text{ m}^2$$

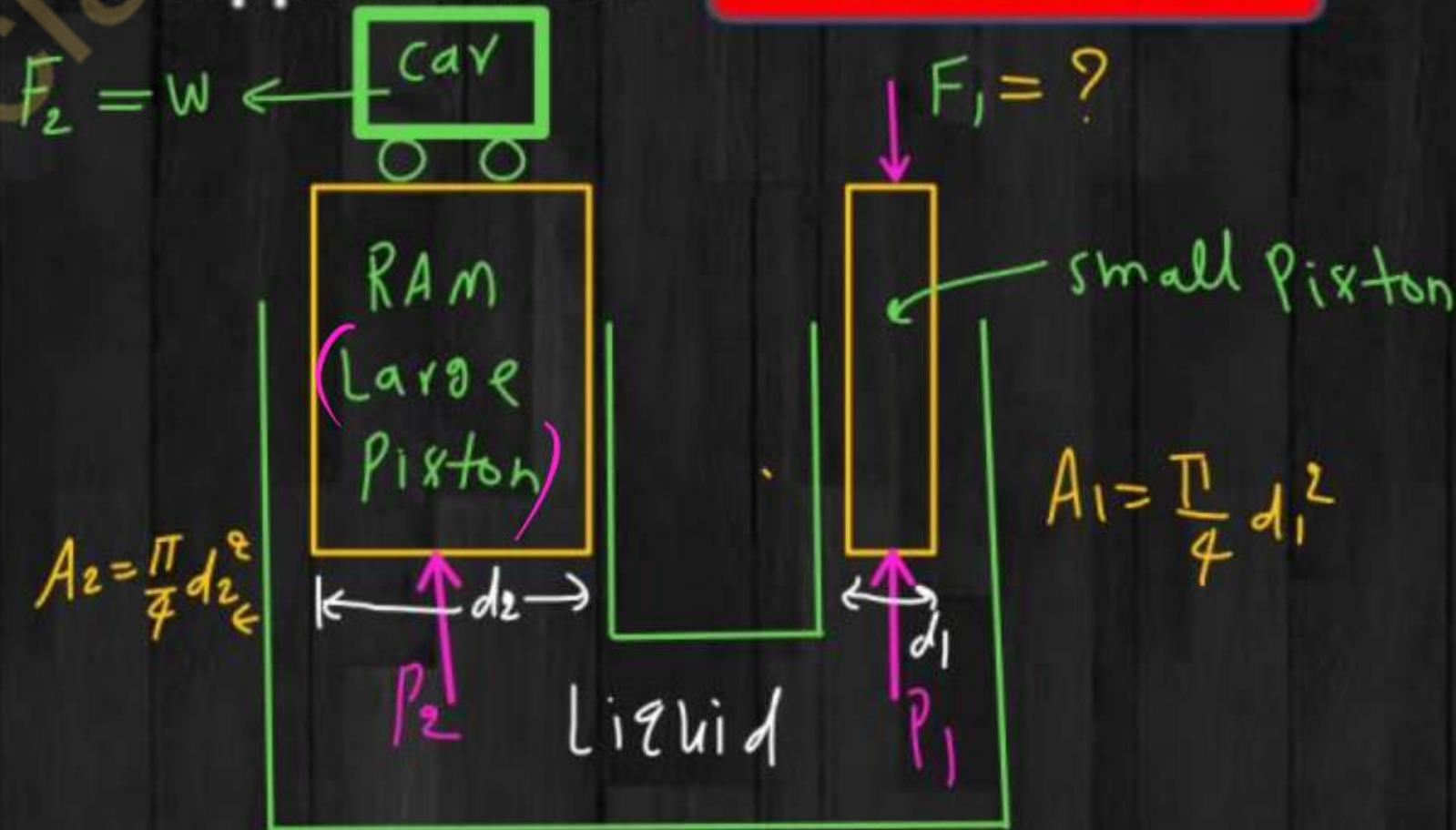
$$F_1 = ?$$

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1 = \frac{A_1}{A_2} F_2$$

$$F_1 = \frac{0.2}{0.9} \times 1.2 \times 10^4$$

$$F_1 = 2666.67 \text{ N} \text{ Ans}$$



Pascal's Law

Numerical.2 A pressure of 2000 Pa is transmitted throughout a liquid column due to a force being applied on a piston. If the piston has an area of 0.1 m^2 , what force is applied.

Given:

$$P = 2000 \text{ Pa (N/m}^2\text{)}$$

$$A = 0.1 \text{ m}^2$$

$$F = ?$$

$$P = \frac{F}{A}$$

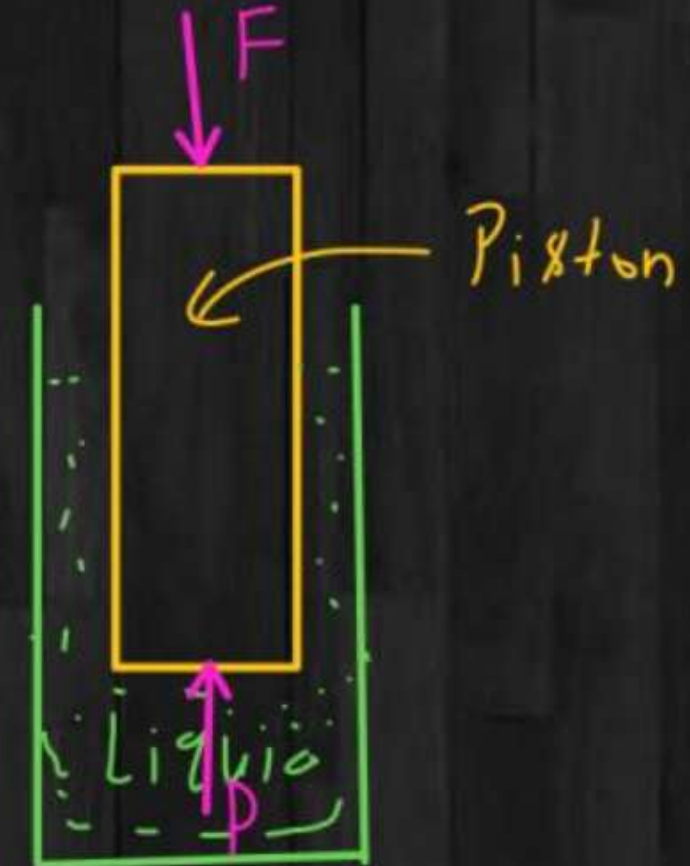
$$F = P \cdot A$$

$$= 2000 \times 0.1 \left(\frac{\text{N}}{\text{m}^2} \times \text{m}^2 \right)$$

$$F = 200 \text{ N}$$

Ans

AKTU : 2023-24 Odd



Pascal's Law

Q.1 Car's weight = 16,000 N, Plunger area = 50 cm^2 , Ram area = 4000 cm^2 . Determine the force required to lift the car. **Ans.** 200 N ✓

Q.2 Two pistons of a hydraulic lift have diameters of 60 cm and 5 cm. What is the force exerted by the larger piston when 50 N is placed on the smaller piston? **Ans.** 7200 N

Q.3 A pressure of 2000 N/m^2 is transmitted throughout a liquid column due to a force being applied on a piston. If the piston has an area of 0.1 m^2 , what force is applied? **Ans.** 200 N

Q.4 A hydraulic car lift can raise vehicles weighing up to 3000 kg. The piston carrying the load has a cross-sectional area of 425 cm^2 . What would be the maximum pressure that the smaller piston would have to handle? **Ans.** $6.917 \times 10^5 \text{ Pa}$ 🐱

CONTINUITY EQUATION

- This equation is based on the “principle of conservation of mass”.
- The quantity of fluid per second is constant at all the cross sections through the pipe.

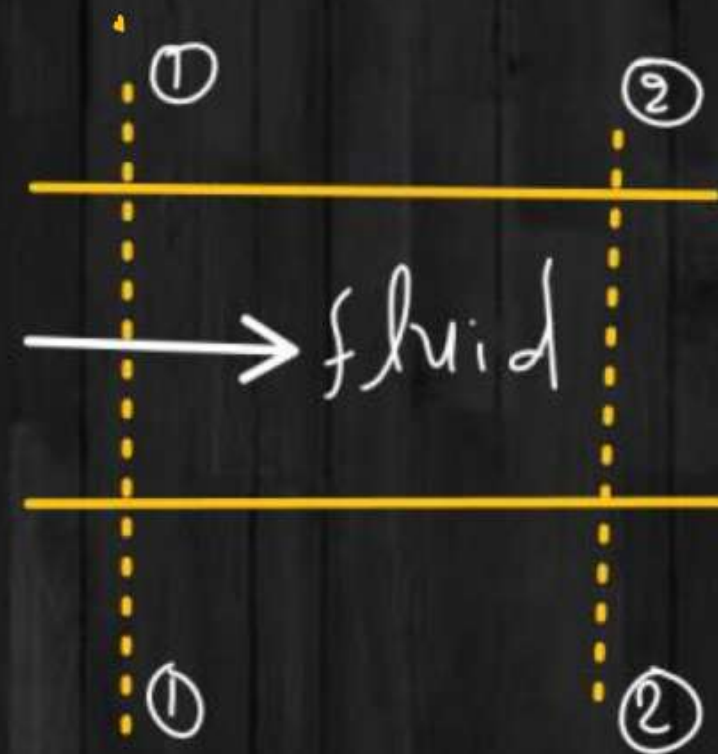
Flow Rate → 1. Volume flow rate = $A \cdot V$ (m^3/s)
(Discharge, Q)

Flow Rate → 2. mass flow rate = $\rho \cdot A \cdot V$ (kg/s)

Where V = velocity

A = cross-sectional area

ρ = density



Consider two cross sections as shown in figure:

Let v_1 = Average velocity at cross-section 1-1 ρ_1 = density at section 1-1

A_1 = Area of pipe at section 1-1

and v_2, ρ_2, A_2 are corresponding values at section 2-2.

Rate of flow at section at 1-1 = $\rho_1 A_1 v_1$

Rate of flow at section at 2-2 = $\rho_2 A_2 v_2$

According to law of conservation of mass:

Rate of flow at section at 1-1 = Rate of flow at section at 2-2

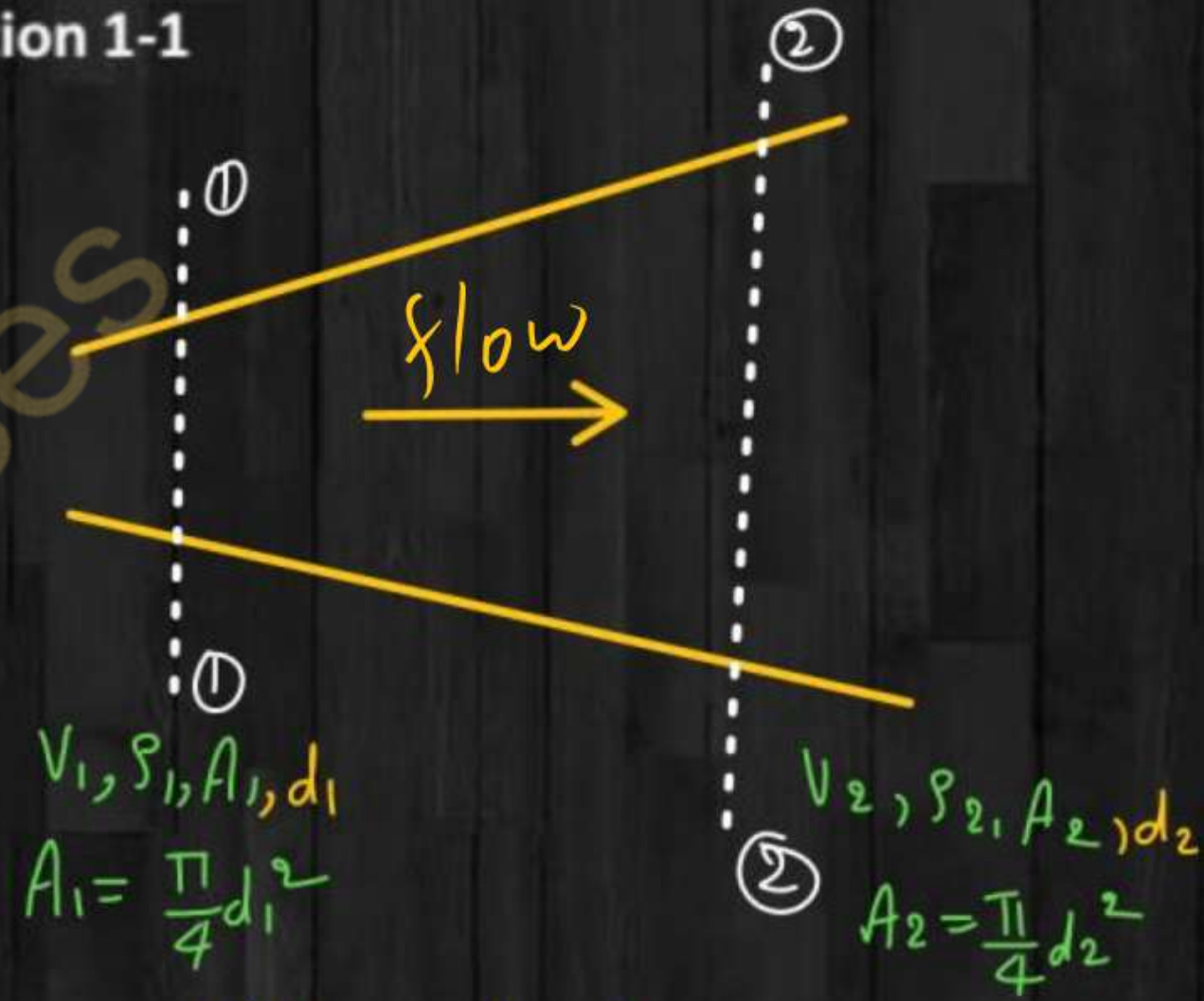
$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

→ Valid for compressible fluid i.e. gases

→ If the fluid is In-compressible, i.e. water then $\rho_1 = \rho_2$ and continuity equation becomes

$$A_1 v_1 = A_2 v_2$$

→ Valid for Incompressible fluid (Liquids)



Numerical-1 Water flows through a pipe of internal diameter 20 cm at the speed of 1 m/s. What should the diameter of the nozzle be if the water is to emerge at the speed of 4 m/s. By continuity equation? [AKTU 2023-24]

given

$$d_1 = 20 \text{ cm}$$

$$V_1 = 1 \text{ m/s}$$

$$V_2 = 4 \text{ m/s}$$

$$d_2 = ?$$

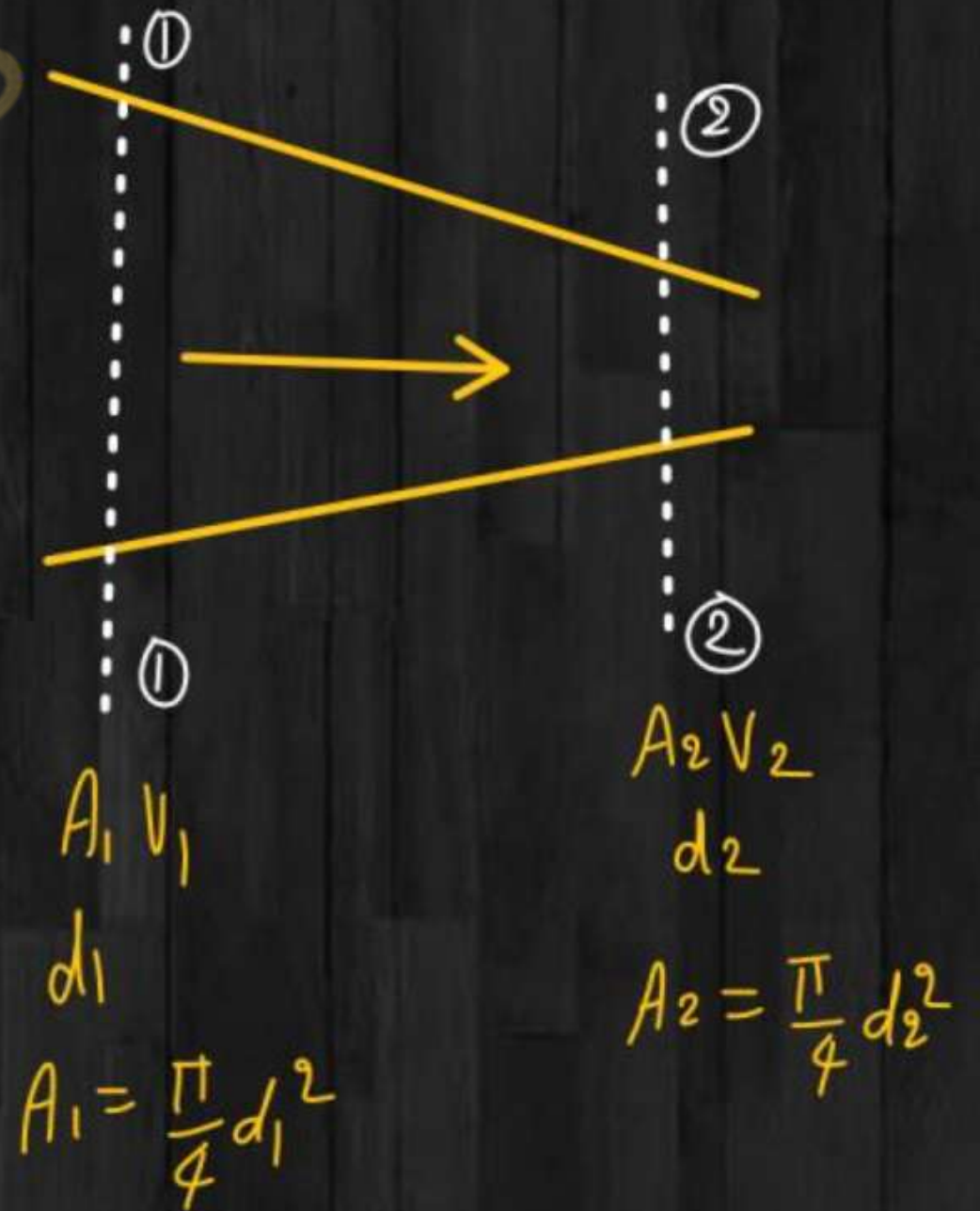
By continuity eqn

$$A_1 V_1 = A_2 V_2$$

$$\frac{\pi}{4} d_1^2 \cdot V_1 = \frac{\pi}{4} d_2^2 \cdot V_2$$

$$(20)^2 \cdot 1 = d_2^2 \cdot 4$$

$$d_2 = 10 \text{ cm} \quad \text{Ans}$$



Numerical-2 The diameters of a pipe at the sections 1 and 2 are 10 cm and 15 cm respectively. Find the discharge through the pipe if the velocity flowing through the pipe at section 1 is 5 m/s. Determine also the velocity at section 2.

Given:

$$d_1 = 10 \text{ cm} \\ = 0.1 \text{ m}$$

$$d_2 = 15 \text{ cm} \\ = 0.15 \text{ m}$$

$$V_1 = 5 \text{ m/s}$$

(i) Discharge = ?

(ii) $V_2 = ?$

$$\text{Discharge} = \text{Volume flow rate (Q)}$$

$$= A_1 V_1 = A_2 V_2$$

$$= \frac{\pi}{4} d_1^2 \times V_1$$

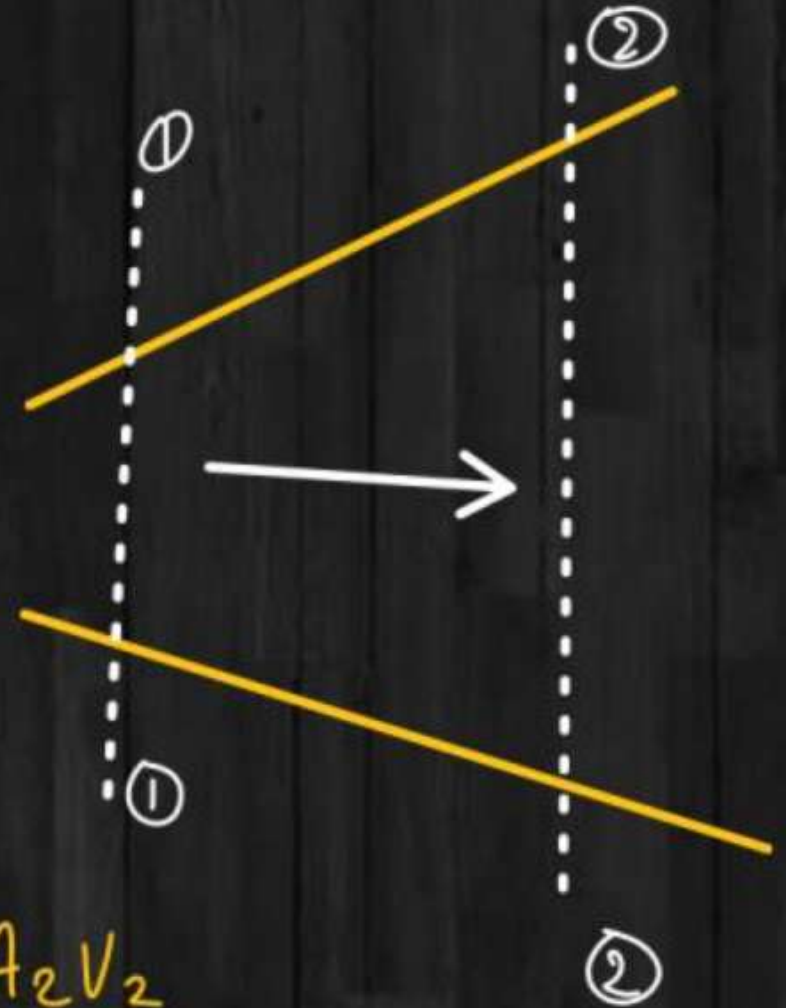
$$= \frac{\pi}{4} (0.1)^2 \times 5$$

$$\boxed{\text{Discharge} = 0.03925 \text{ m}^3/\text{s}} \text{ Ans}$$

$$\text{Discharge} = A_1 V_1 = 0.03925 = A_2 V_2$$

$$= \frac{\pi}{4} d_2^2 \times V_2 = \frac{\pi}{4} (0.15)^2 \times V_2$$

$$\Rightarrow \boxed{V_2 = 2.22 \text{ m/s}} \text{ Ans}$$



DPP

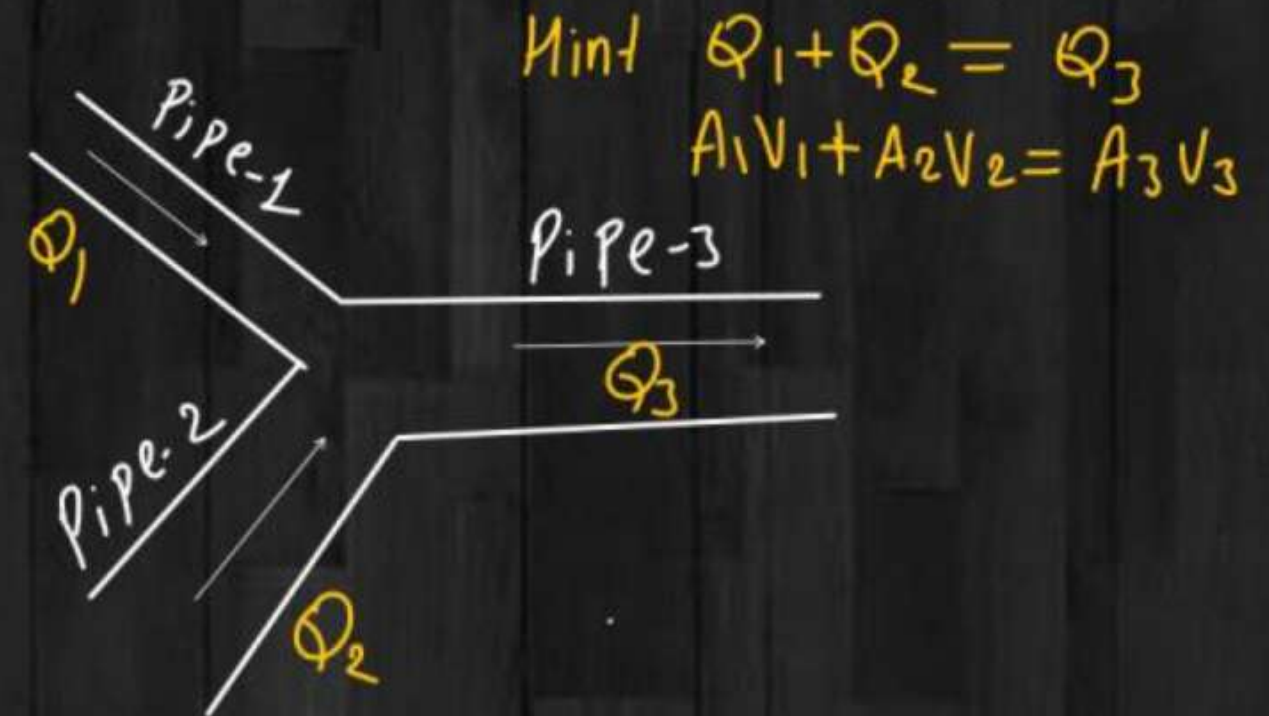
Numerical-1 A liquid flows through a pipe with a diameter of 10cm at a velocity of 9cm/s. If the diameter of the pipe then decreases to 6cm, what is the new velocity of the liquid?

Ans. 25cm/s

Numerical-2 Consider steady flow of water in a situation where two pipe lines (pipe 1 and pipe 2) combine into a single pipe line (line 3) as shown in figure. The cross-sectional area of all three pipelines are constant. The following data is given.

Pipe Number	Area(m ²)	Velocity(m/s)
1	1 A_1	1 V_1
2	2 A_2	2 V_2
3	2.5 A_3	? $V_3 = ?$

Determine velocity in pipe-3.



Hydraulic Machines

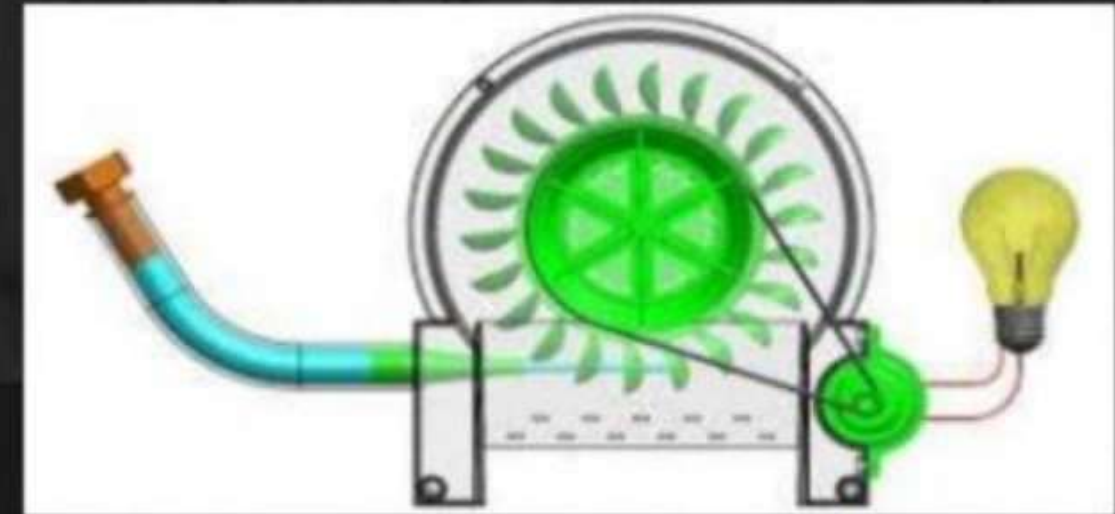
→ Hydraulic machines are defined as those machines which convert either **hydraulic energy into mechanical energy** or **mechanical energy into hydraulic energy**.

→ **Hydraulic Energy**- Energy possessed by water.

→ **Mechanical Energy**- power produced at shaft of the turbine.

Turbines- Those hydraulic machines which convert the hydraulic energy into mechanical energy, are called as turbines. $HE \rightarrow ME$

Pumps- Those hydraulic machines which converts mechanical energy into hydraulic energy, are called as pump. $ME \rightarrow HE$



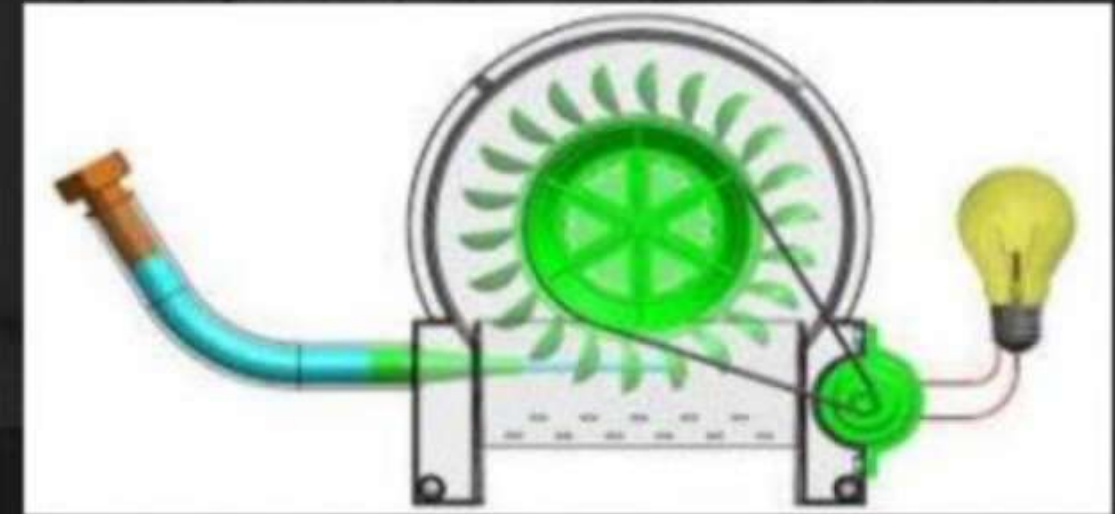
Turbine

→ Turbines are defined as the machines which convert the hydraulic energy into mechanical energy.

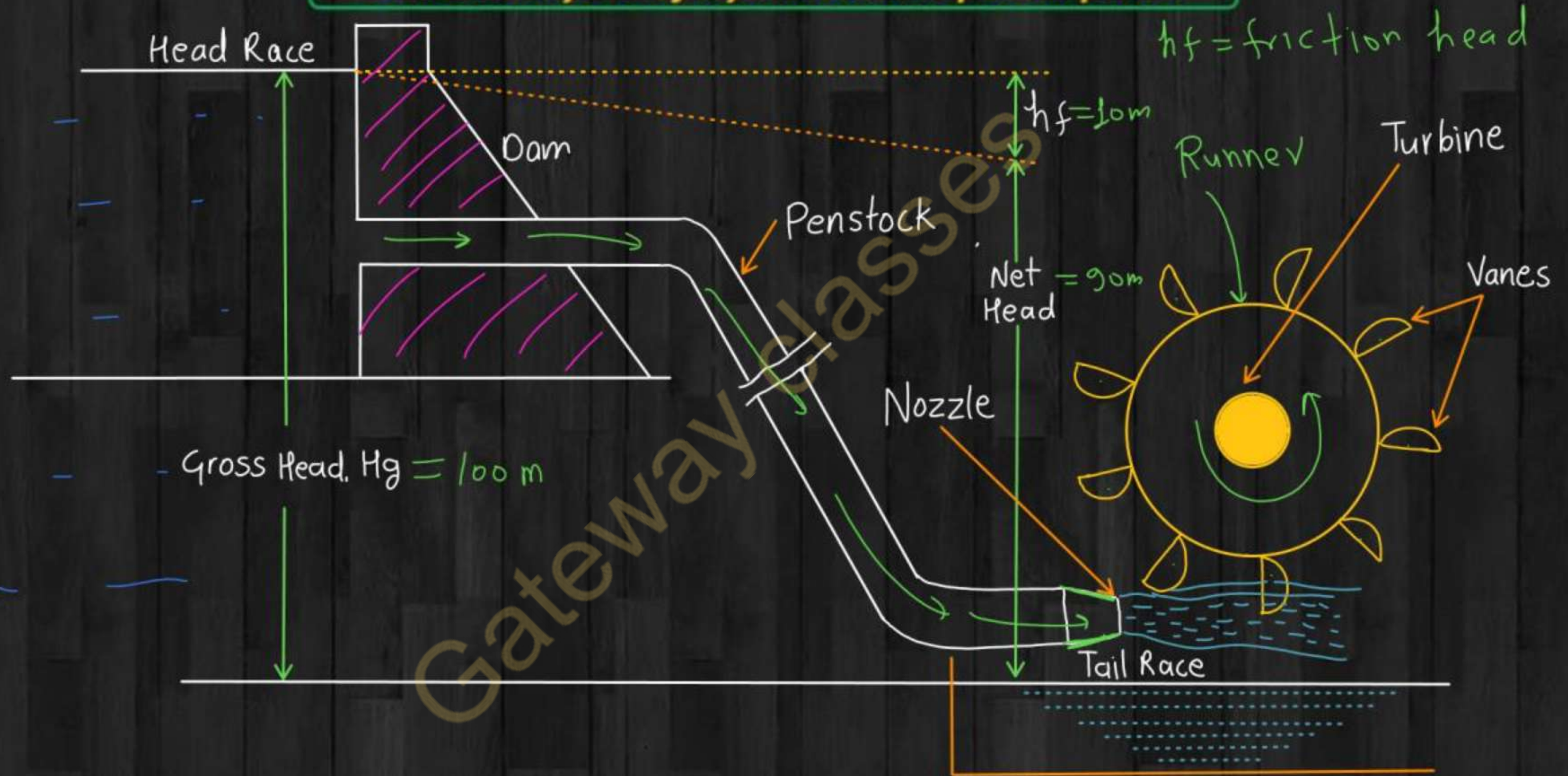
→ This mechanical energy is used in running an electric generator which is directly coupled to the shaft of the turbine.

→ Then the mechanical energy is converted into electrical energy.

→ The electric power which is obtained from hydraulic energy is known as *Hydro-electric power*.



General Layout of Hydro-electric power plant



Classification of turbine

The turbines are classified in the following ways:-

1. According to the type of energy available at inlet of a turbine

✓ (a) Impulse turbine

✓ (b) Reaction turbine

→ If at the inlet of the turbine, **only kinetic energy** is available, the turbine is known as an *impulse turbine*.

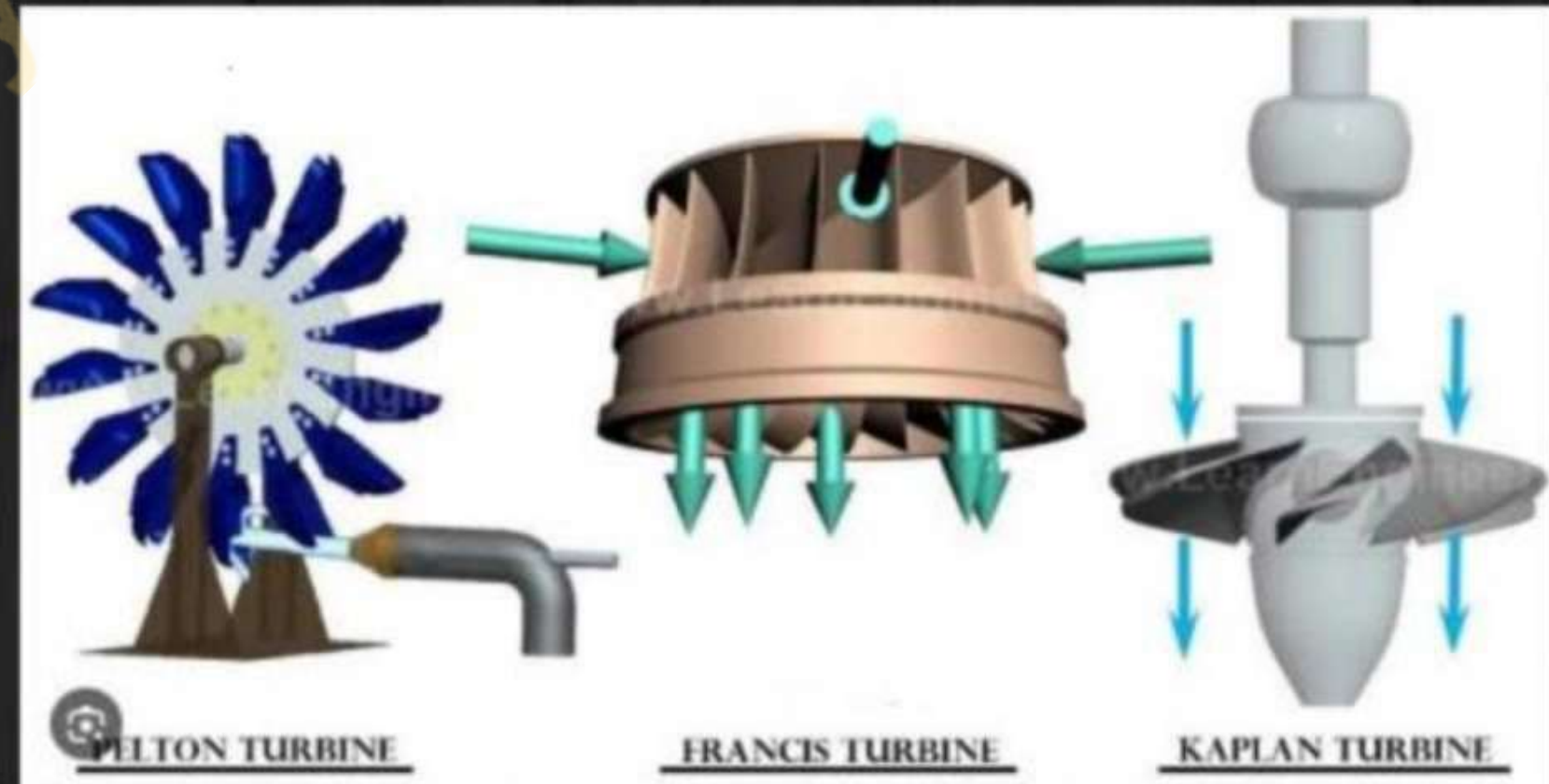
Example-Pelton Turbine

→ If at the inlet of the turbine, water possesses **kinetic energy as well as pressure energy**, the turbine is known as a *reaction turbine*.

Examples-Francis turbine, and Kaplan turbine.

2. According to the *direction of flow of water through runner*:

- a) Tangential flow turbine (e.g. Pelton Turbine)
- b) Radial flow turbine (e.g. Francis Turbine)
- c) Axial flow turbine (e.g. Kaplan Turbine)
- d) Mixed flow turbine (e.g. Modern Francis Turbine)



3. According to head available at the inlet of turbine

- a) High head turbine (e.g. Pelton Turbine)
- b) Medium head turbine (e.g. Francis Turbine)
- c) Low head turbine (e.g. Kaplan Turbine)

4. According to the specific speed of turbine

- a) Low specific speed turbine (e.g. Pelton Turbine)
- b) Medium specific speed turbine (e.g. Francis Turbine)
- c) High specific speed turbine (e.g. Kaplan Turbine)

5. According to the position of shaft of turbine

- a) Horizontal shaft turbine
- b) Vertical shaft turbine

(Pelton turbine has horizontal shaft whereas the rest have vertical shaft)

6. According to name of originator

- a) Pelton Turbine(Pelton Wheel)
- b) Francis Turbine
- c) Kaplan Turbine

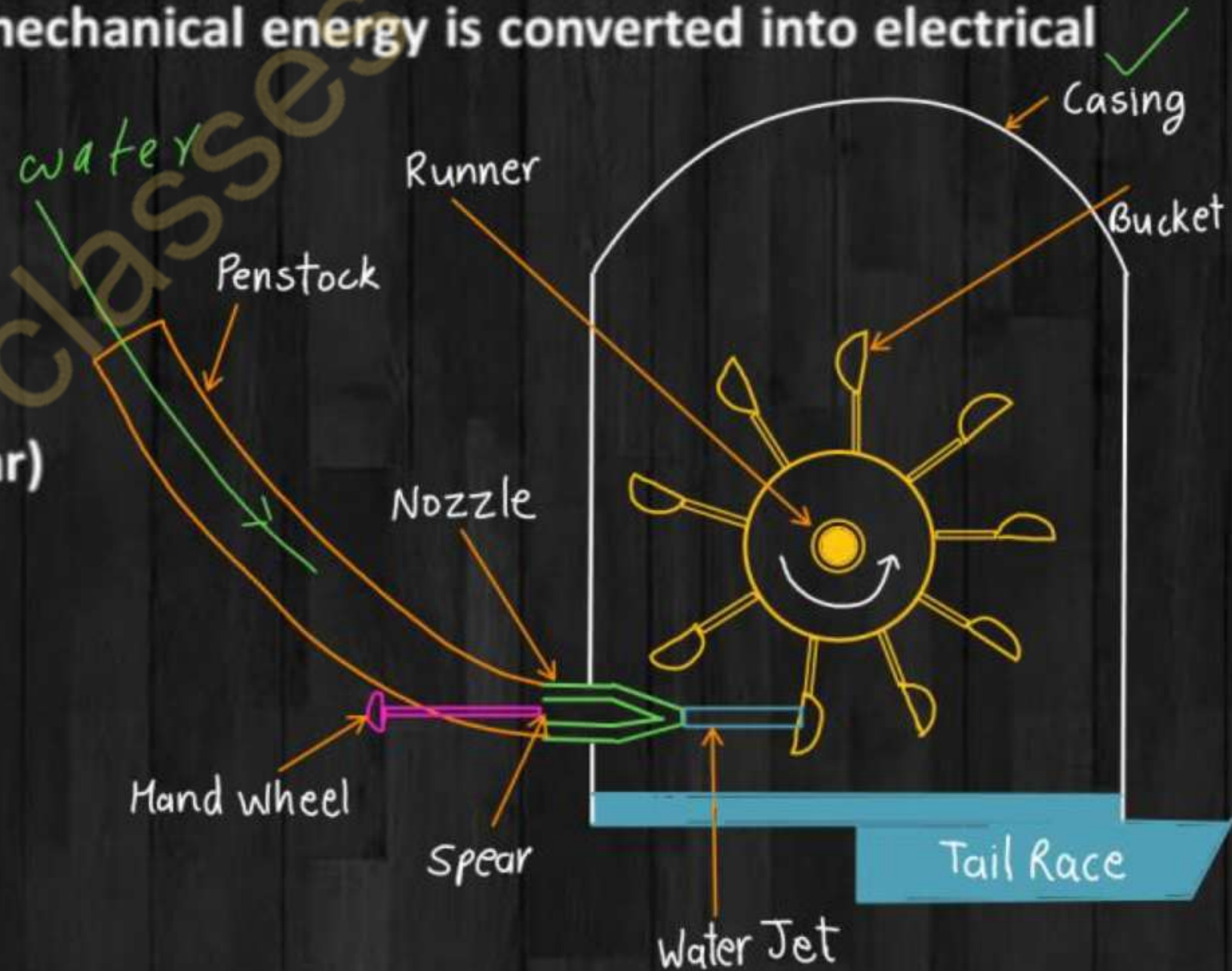
Impulse Turbine (Pelton)

RGPV / AKTU

The principle of a Pelton impulse turbine is based on converting the kinetic energy of a high-velocity water jet into mechanical energy then mechanical energy is converted into electrical energy with the help of generator.

Main parts of Impulse turbine (Pelton) are :-

1. Nozzle and flow regulating Arrangement (Spear)
2. Runner and buckets
3. Casing
4. Breaking jet



1. Nozzle and Flow Regulating Arrangement (Spear)

- Nozzle is used to increase the kinetic energy of the water that is going to strike the buckets or vanes attached to the runner.
- The quantity of water that strikes the buckets is controlled by a spear. The spear is installed inside the nozzle and regulates the flow of water that is going to strike on the vanes of the runner. A nozzle containing a spear is shown in the figure.
- When the spear is move backward the rate of flow of water increases and when it is pushed forward the rate of flow of water decreases.

2. Runner and Buckets

- A wheel of the turbine consist of series of buckets/blades/vanes mounted on its periphery.
- The buckets are designed in such a way that the jet of water strikes the buckets, deflected through 160 degree to 170 degree.
- The buckets of the Pelton turbine are made up of **cast iron, cast steel bronze or stainless steel.**

3. Casing: The outer covering of this turbine is called casing. The Pelton turbine with the casing is shown in the figure.

- It is used to avoid accidents and prevents the splashing of water.
- It does not perform any hydraulic function.
- The pressure throughout the turbine from inlet to outlet is atmospheric in case impulse turbine.
- **Cast iron or fabricated steel plates** are used to make the casing of the Pelton Turbine.

4. Breaking jet

- When the jet of water is completely closed by pushing the spear in the forward direction then the amount of water striking the runner becomes zero.
- But still, the runner keeps moving due to the inertia of the runner.
- In order to stop the runner in the shortest possible time, a small nozzle is provided which directs the jet of water at the back of the vanes.
- This jet of water used to stop the runner of the turbine is called breaking jet.

Applications of Pelton Turbine

High-Head Hydroelectric Plants:

→ Efficient for high elevation drops and low water flow.

Mountainous Regions:

→ Suitable for generating power from high-altitude water sources.

Remote Power Systems:

→ Ideal for off-grid electricity generation in remote areas.

Small-Scale Hydropower:

→ Used in micro and mini hydro projects for local power supply.

Industrial Use:

→ Generates electricity in industries with high-head water supply.

Pumped Storage:

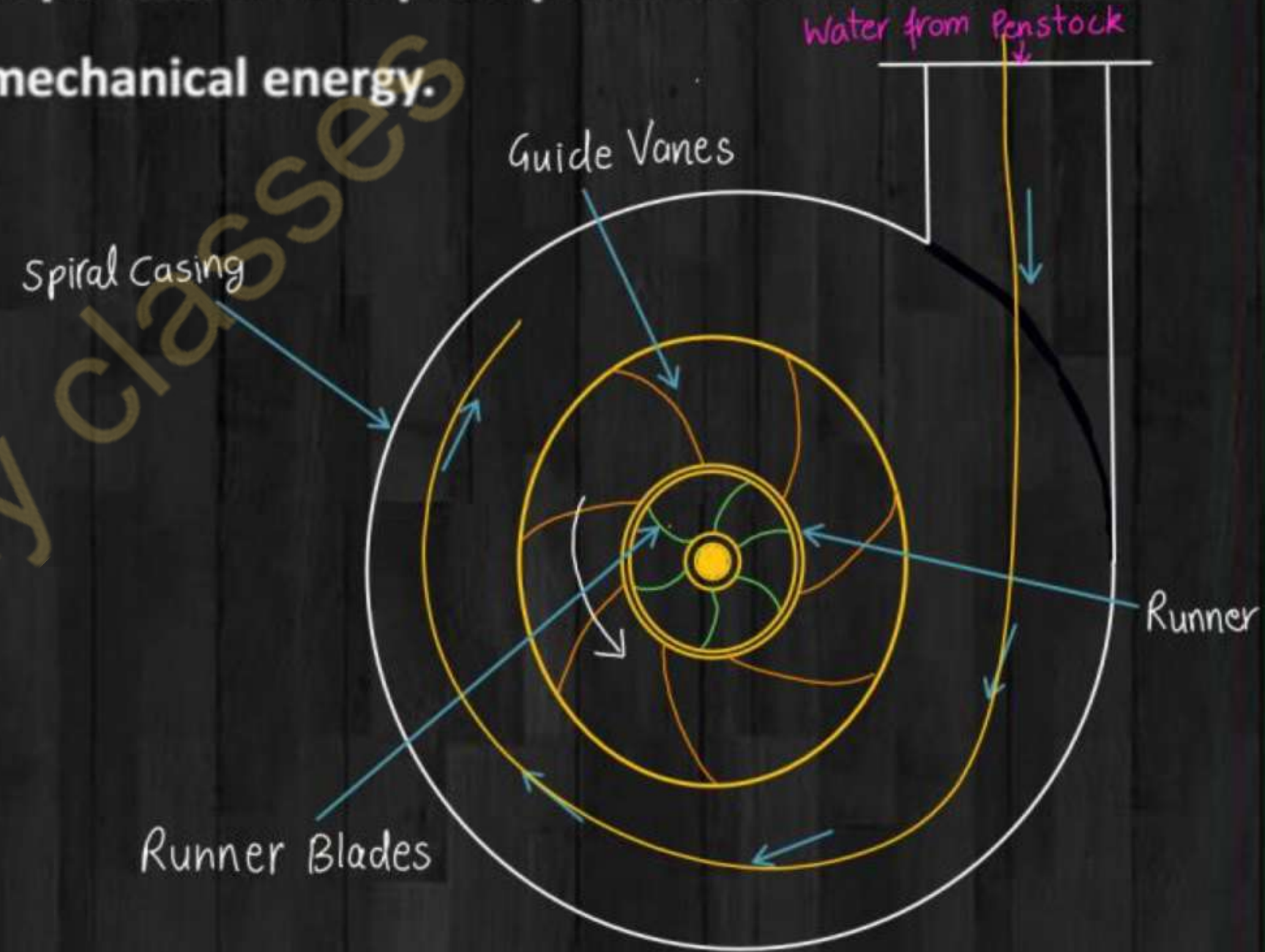
→ Utilized in systems for energy storage and peak power generation.

Reaction Turbine(Francis)

A reaction turbine, specifically a Francis turbine, operates on the principle of both reaction and impulse forces to convert hydraulic energy into mechanical energy.

Main parts of reaction turbine are :-

1. Spiral Casing
2. Guide mechanism (guide vanes)
3. Runner
4. Draft tube



1. Spiral Casing

The spiral casing in a Francis turbine has the following functions:

Uniform Distribution

→ It evenly distributes water around the circumference of the turbine runner to ensure that water flows smoothly and uniformly into the guide vanes and runner blades.

Pressure Maintenance:

→ It helps maintain the pressure of the water as it spirals towards the runner, ensuring that energy is effectively transferred to the turbine blades.

Flow Direction:

→ The spiral shape directs the water flow in a way that optimizes the angle at which water hits the runner blades, maximizing efficiency.

Structural Support:

→ It provides structural support and houses the guide vanes and other components, ensuring that they are correctly aligned and function effectively.

2. Guide Vanes

- The guide vanes allow the water to strike the fixed blades on the runner without shock at inlet.
- Guide vanes are not stationary, they change their angle as per the requirement to control the angle of striking of water to turbine blades to increase the efficiency.
- They also regulate the flow rate of water into the runner blades thus controlling the power output of a turbine according to the load on the turbine.

3. Runner Blades

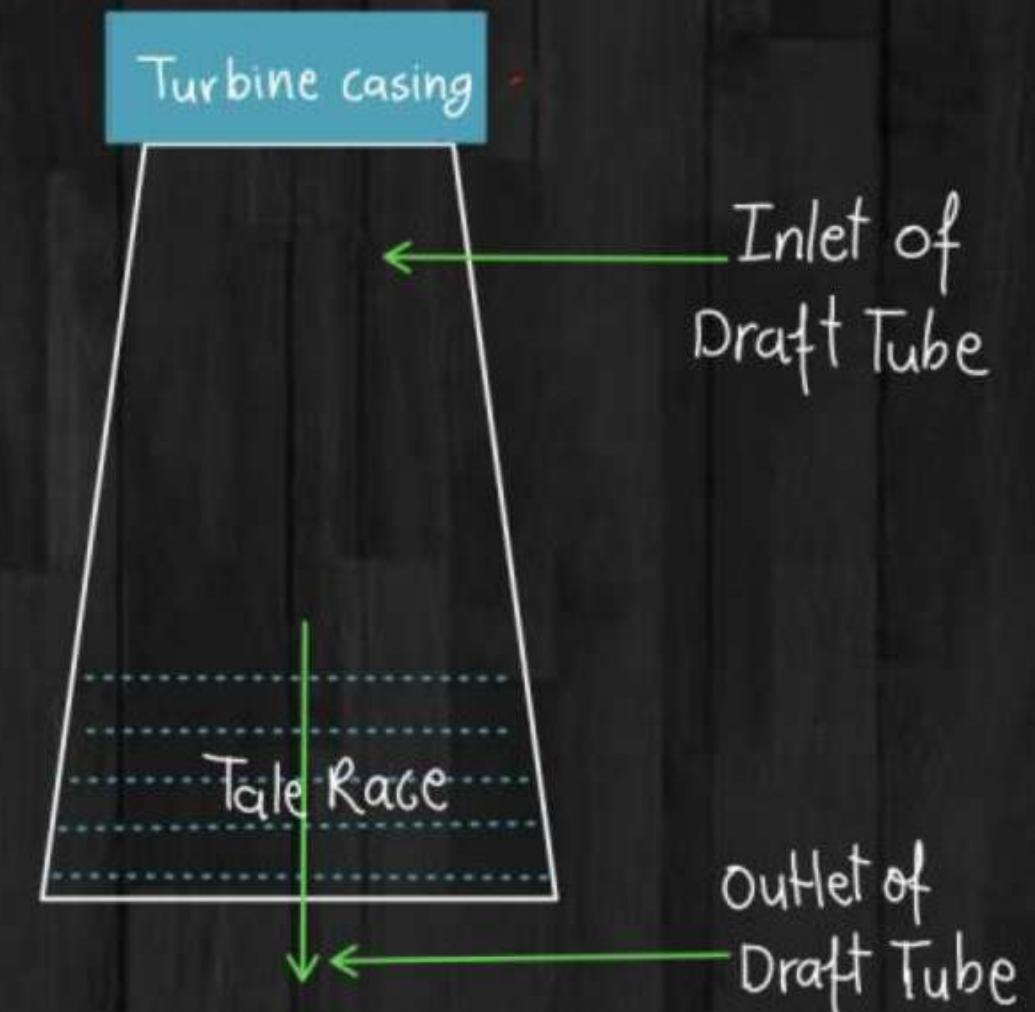
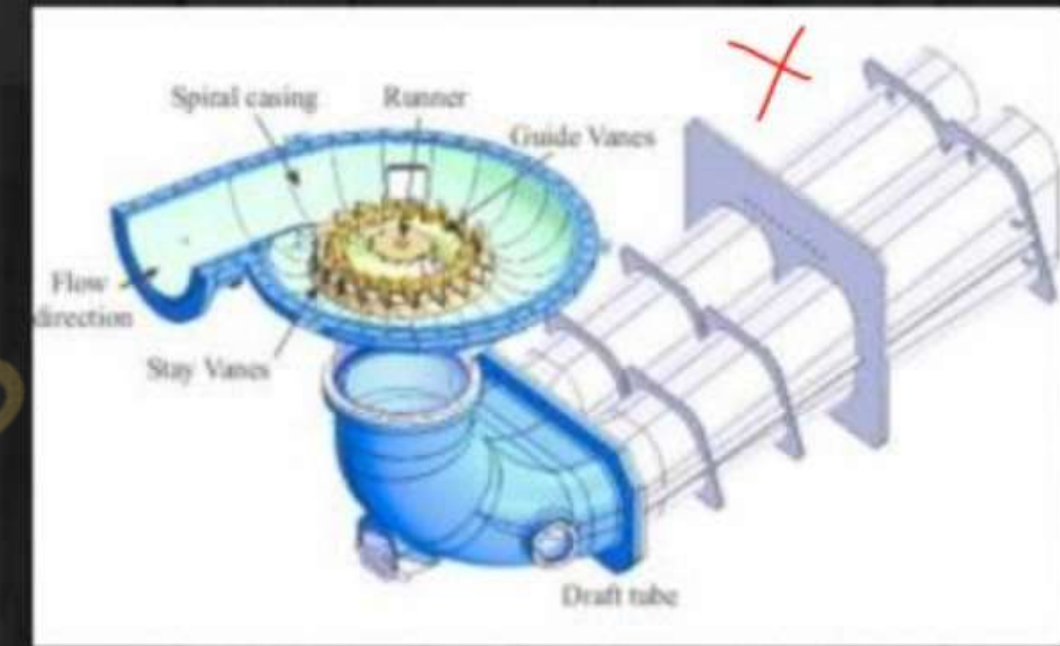
- These are the fixed blades mounted on the periphery of the runner.
- The performance and efficiency of the turbine dependent on the design of the runner blades.

4. Draft tube-

→ Draft Tube is a diverging tube fitted at the exit of the runner of the turbine and used to utilize the kinetic energy available with water at the exit of the runner.

→ Pressure head is increased by decreasing the exit velocity.

→ Overall efficiency and the output of the turbine can be improved.



Applications of Francis Turbine

Hydroelectric Power Plants:

→ Used in medium to large-scale plants for efficient power generation.

Pumped Storage Plants:

→ Employed for energy storage and peak power generation.

Irrigation Systems:

→ Generates electricity while aiding in water distribution.

Water Supply Systems:

→ Produces power from water flow in pipelines.

Run-of-the-River Projects:

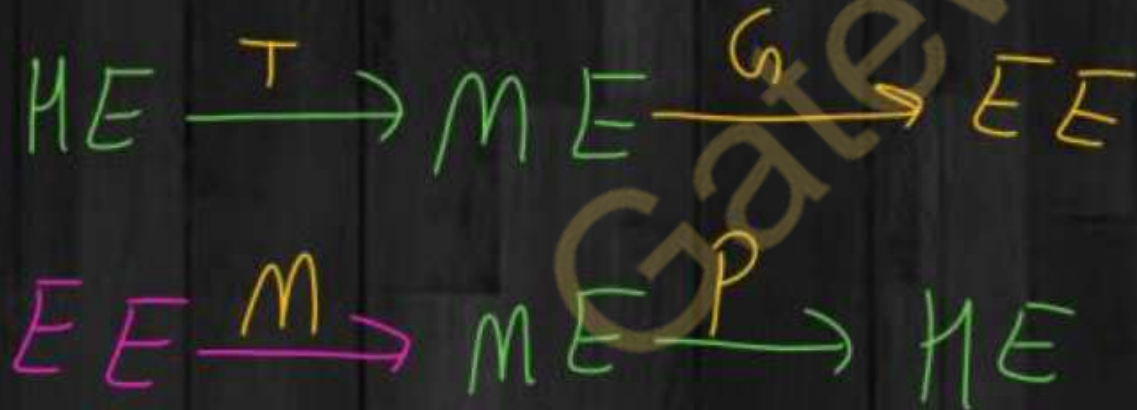
→ Suitable for generating power without large reservoirs.

Industrial Applications:

→ Used in industries to generate electricity from process water flows.

Pump

- The hydraulic machine which converts *mechanical energy into hydraulic energy* is known as a pump.
- The hydraulic energy is in the form of *Pressure Energy*.
- If the mechanical energy is converted into pressure energy by means of centrifugal force acting on the fluid, the hydraulic machine is called *Centrifugal Pump*.
- The centrifugal pump works on the principle of *forced vortex flow*.



Classification of hydraulic Pump

→ There exist a wide variety of pumps that are designed for various specific applications.

→ However, most of them can be broadly classified into two categories as mentioned below-

1. Dynamic Pressure Pumps ✓
2. Positive Displacement Pump ✓

Non-positive displacement

Dynamic Pressure
Pumps

Centrifugal ✓

Axial flow

Radial flow

Positive displacement

Rotary

Gear

Vane

Screw

Reciprocating ✓

Axial piston

Radial piston

Centrifugal Pump

[A K T U]

The principle of a centrifugal pump is based on converting rotational kinetic energy, typically from an electric motor or engine, into hydrodynamic energy of fluid flow.

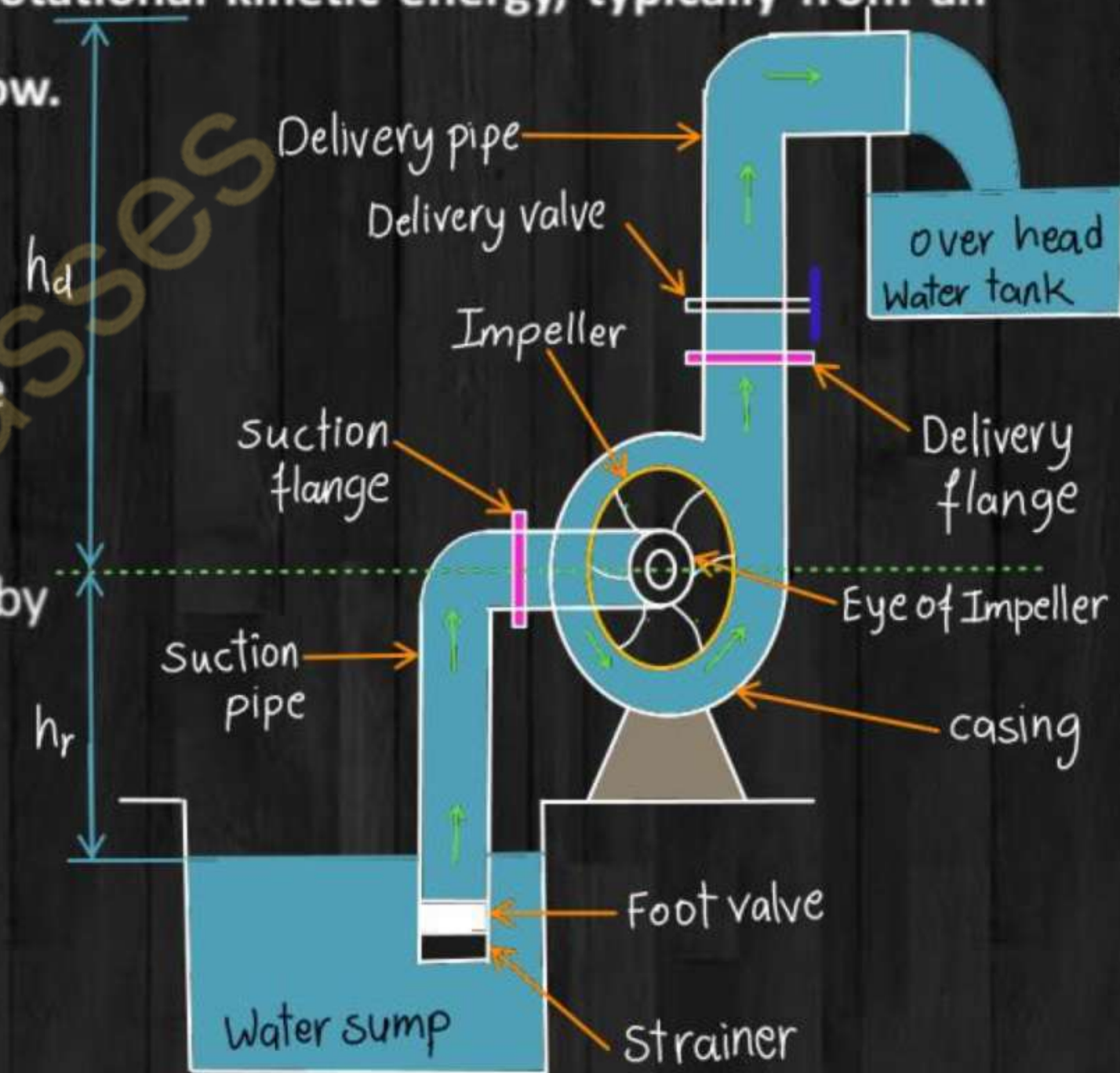
Main parts of a Centrifugal Pump are :-

1. Impeller ✓

- The rotating part of the pump that transfers energy to the fluid.
- Typically has vanes or blades that push the fluid outward by centrifugal force.

2. Pump Casing ✓

- The outer shell that houses the impeller.
- It converts kinetic energy from the impeller into pressure energy.
- Directs the fluid to the discharge nozzle.



3. Suction and Delivery Pipes

Suction Pipe: Carries fluid from the source (e.g., water sump).

Delivery Pipe : Carries the pressurized fluid to the delivery point (e.g., overhead water tank).

4. Shaft

- Connects the impeller to the motor.
- Transfers rotational energy from the motor to the impeller.

5. Seal or Packing

- Prevents fluid from leaking along the shaft where it passes through the pump casing.
- Can be a mechanical seal or packing material.

6. Bearings

- Support the shaft and allow it to rotate smoothly.
- Absorb radial and axial loads from the impeller.

7. Motor

- Provides the power to rotate the impeller.
- Can be electric or driven by an engine.

Working of a Centrifugal Pump

Fluid Entry

- Fluid enters the pump through the suction pipe, which is connected to the suction nozzle.
- The suction nozzle is centrally located to the impeller's eye, ensuring even flow distribution.

Impeller Action

- The motor drives the shaft, causing the impeller to rotate.
- As the impeller spins, it imparts kinetic energy to the fluid, accelerating it outward radially.

Centrifugal Force

- The centrifugal force generated by the rotating impeller causes the fluid to move from the impeller's eye to its periphery.
- This outward movement increases the fluid's velocity and kinetic energy

Energy Conversion in Casing

- The high-velocity fluid enters the pump casing or volute, where the cross-sectional area increases.
- This gradual expansion reduces the fluid's velocity and converts kinetic energy into pressure energy.

Discharge

- The pressurized fluid is directed towards the discharge nozzle.
- It exits the pump at a higher pressure, ready for distribution to the desired location (e.g., an overhead water tank).

Continuous Operation

- The process continues as long as the impeller rotates and there is a fluid supply.
- This continuous movement ensures a steady flow and pressure of the fluid being pumped.

Applications of Centrifugal Pumps

→ Centrifugal pumps are the most popular choice for the discharge of fluid.

→ These are the following applications of Centrifugal Pumps:-

- ✓ **1. Oil & Energy** - pumping crude oil, slurry, mud; used by refineries, power generation plants
- ✓ **2. Industrial & Fire Protection Industry** - Heating and ventilation, boiler feed applications, air conditioning, pressure boosting, fire protection sprinkler systems.
- ✓ **3. Waste Management, Agriculture & Manufacturing** - Wastewater processing plants, municipal industry, drainage, gas processing, irrigation, and flood protection
- ✓ **4. Pharmaceutical, Chemical & Food Industries** - paints, hydrocarbons, petrochemical, cellulose, sugar refining, food and beverage production
- ✓ **5. Various industries (Manufacturing, Industrial, Chemicals, Pharmaceutical, Food Production, Aerospace etc.)** - for the purposes of cryogenics and refrigerants.

Reciprocating Pump

[AKTU]

→ Reciprocating pump operates by using a piston or plunger to move fluid through a cylinder, creating pressure and flow.

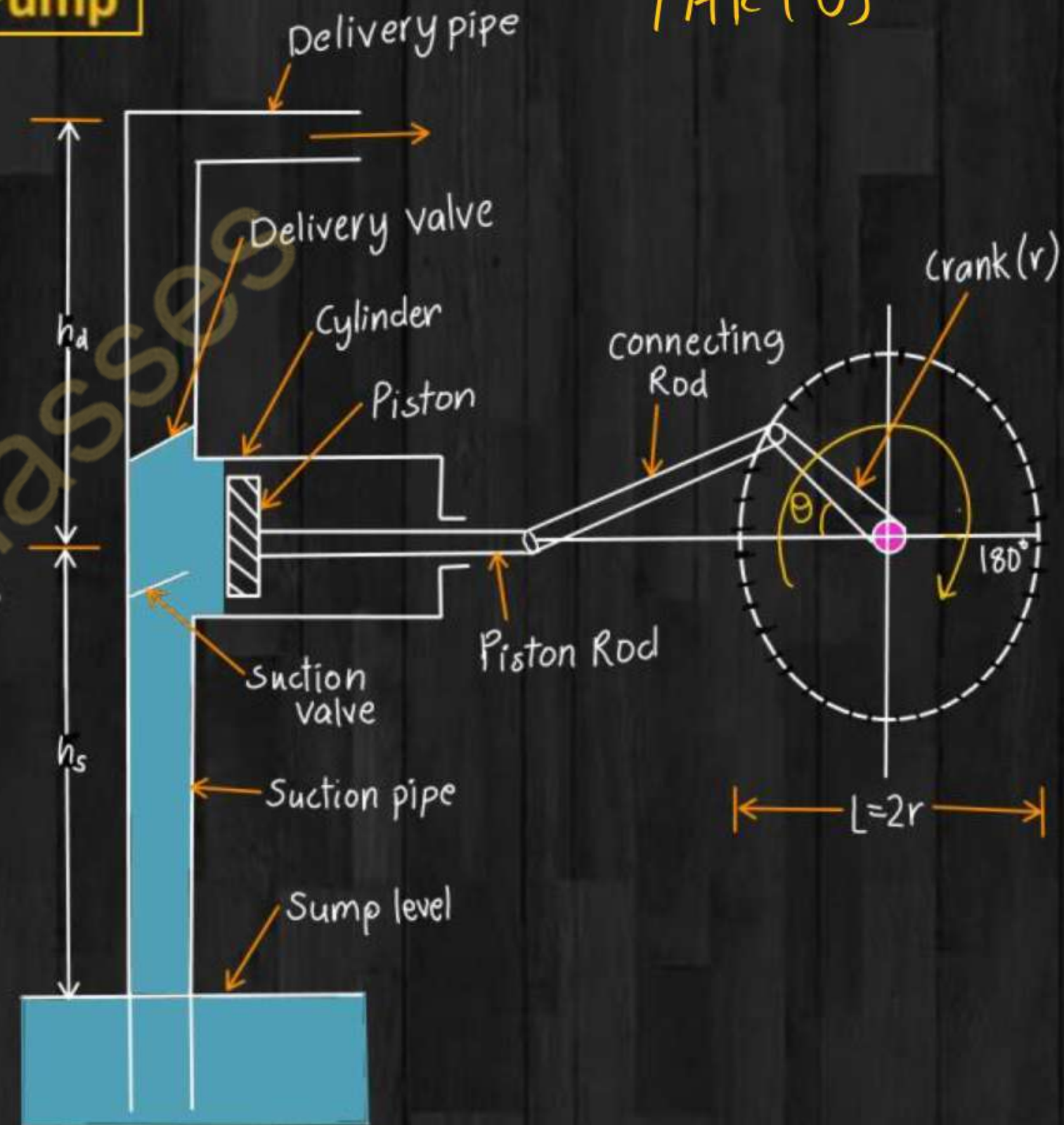
Main Parts of a Reciprocating Pump

1. Cylinder

- A chamber where the piston moves backward and forward.
- Houses the fluid during the pumping process.

1. Piston (or Plunger)

- Moves back and forth within the cylinder.
- Creates a vacuum during the suction stroke and exerts pressure during the discharge stroke.



3. Piston Rod

- Connect the piston to the connecting rod.
- Transmits the motion from the crankshaft to the piston.

4. Connecting Rod

- Links the piston rod to the crankshaft.
- Converts the rotational motion of the crankshaft into the linear motion of the piston.

5. Crankshaft

- Driven by a motor, it rotates to move the connecting rod and piston.
- Converts rotational energy into reciprocating motion.

6. Suction Valve

- Allows fluid to enter the cylinder during the suction stroke.
- Closes during the discharge stroke to prevent backflow.

7. Discharge Valve

- Opens to allow fluid to exit the cylinder during the discharge stroke.
- Closes during the suction stroke to maintain pressure.

8. Suction Pipe

- Connects the fluid source to the suction valve.
- Guides the fluid into the cylinder.

9. Discharge Pipe

- Connects the discharge valve to the output system.
- Directs the pressurized fluid to its destination.

Working of a Reciprocating Pump

Suction Stroke

- The piston moves backward, creating a vacuum inside the cylinder.
- The suction valve opens due to the vacuum, allowing fluid to enter the cylinder from the suction pipe.
- The discharge valve remains closed to maintain pressure.

Discharge Stroke

- The piston moves forward, compressing the fluid in the cylinder.
- The suction valve closes to prevent fluid from flowing back into the suction pipe.
- The discharge valve opens due to the pressure, allowing the fluid to exit the cylinder and flow into the discharge pipe.

Continuous Operation

- The crankshaft continues to rotate, causing the piston to reciprocate back and forth.
- This continuous motion maintains a steady flow of fluid from the suction source to the discharge destination.

Types of Reciprocating Pump

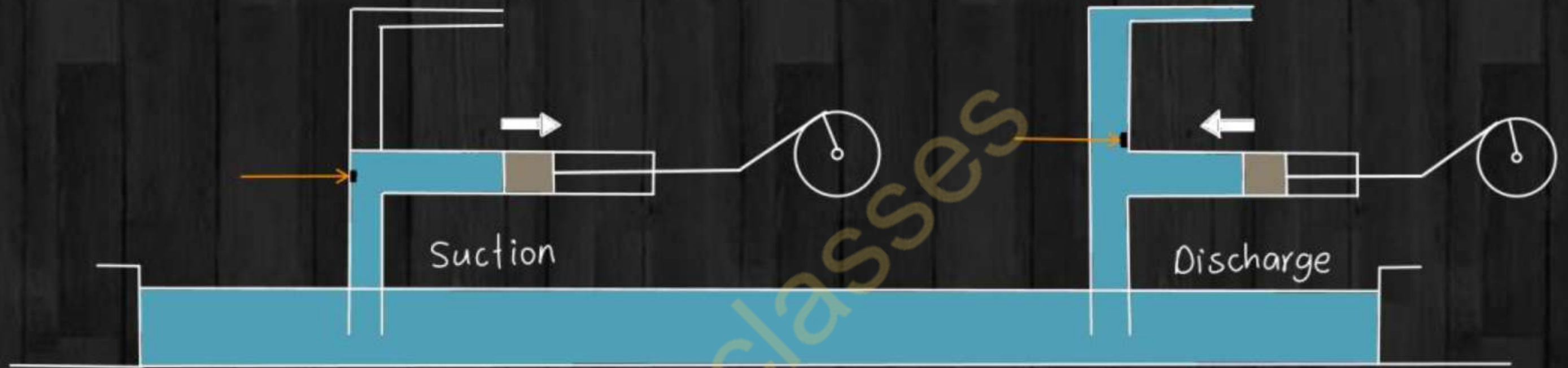
1. According to the water being in contact with one side or both sides of the piston-

- A. Single acting pump
- B. Double acting pump

2. According to the number of cylinder provided

- A. Single-cylinder pump,
- B. Double cylinder pump,
- C. Triple cylinder pump.

A. Single acting pump



A. Double acting pump



Hydraulic Lift

→ A hydraulic lift operates based on Pascal's principle, which states that pressure exerted on a fluid in a closed system is transmitted equally in all directions.

The Hydraulic Lifts are of two types -

1. Direct-acting hydraulic lift
2. Suspended hydraulic lift



1. Direct Acting Hydraulic Lift

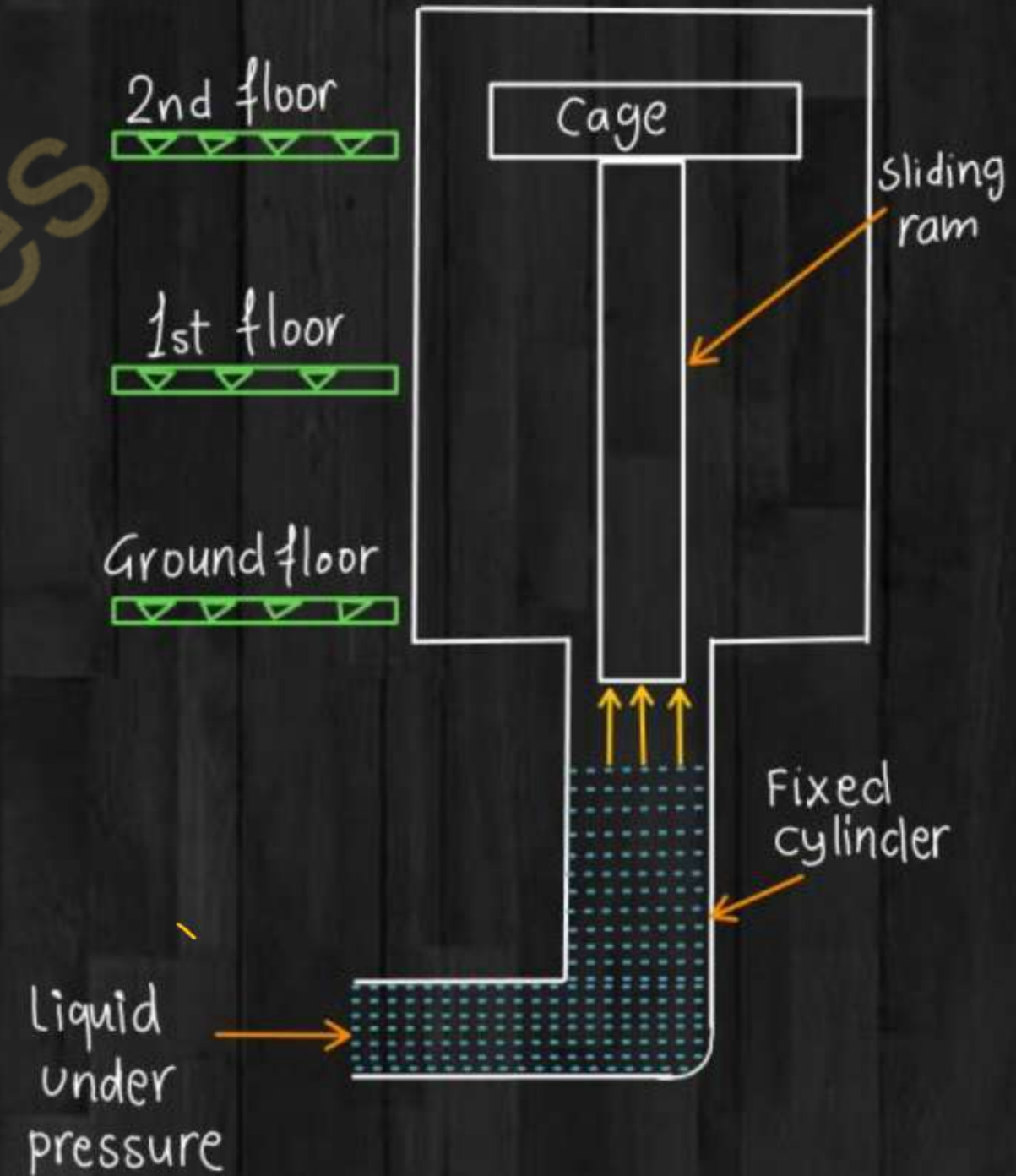
Basic Components

Cage:- The platform that carries the load (e.g., people or goods).

Sliding Ram:- A movable rod or piston connected to the cage.

Fixed Cylinder sealed cylinder filled with hydraulic fluid (usually oil).

Liquid Under Pressure:- The hydraulic fluid that transmits force.



Working

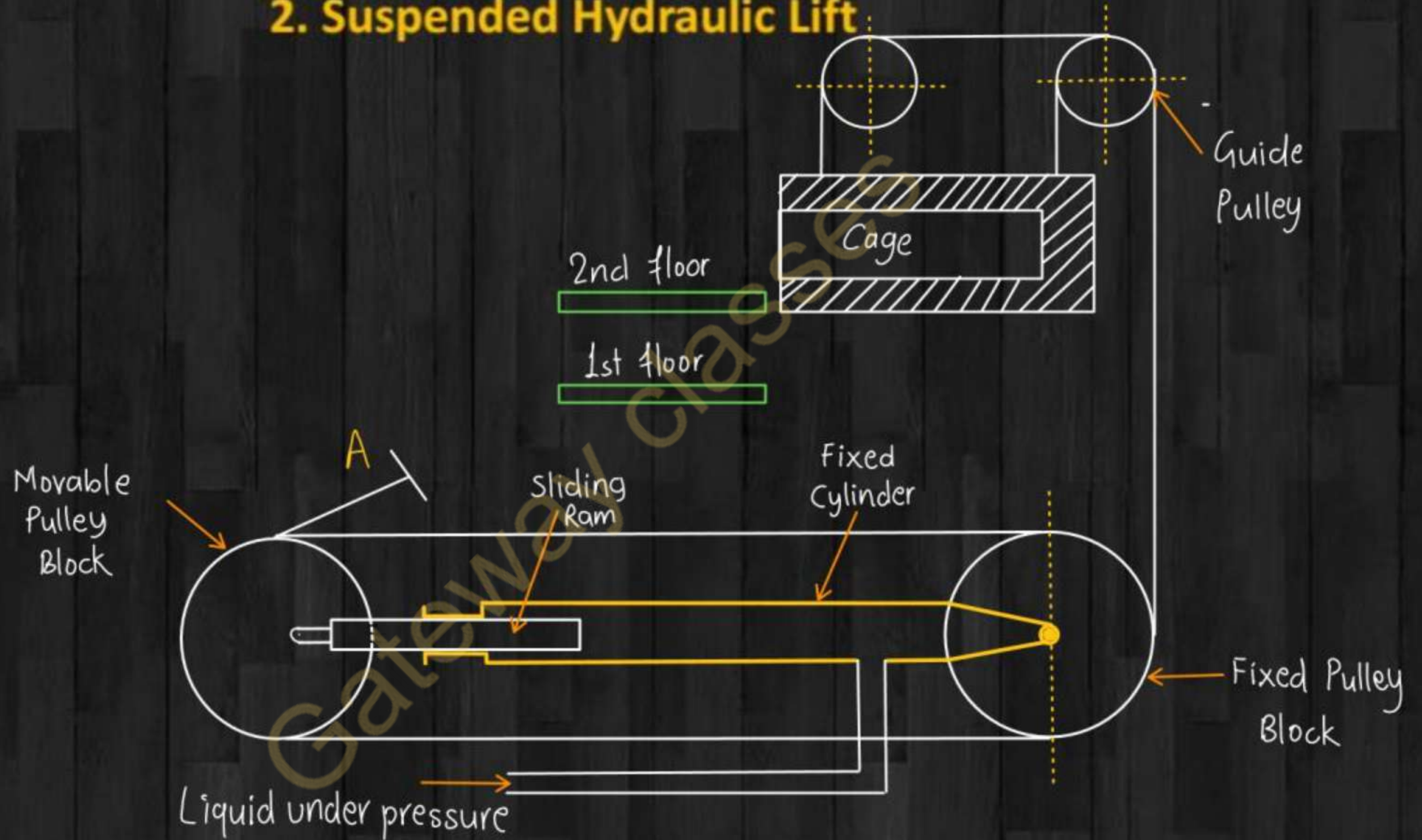
Ground Floor to Upper Floors:

- When the liquid (hydraulic fluid) under pressure is forced into the fixed cylinder, it pushes the sliding ram upwards.
- As the sliding ram rises, it lifts the cage, moving it from the ground floor to the desired upper floors.

Upper Floors to Ground Floor:

- To lower the cage, the pressure on the hydraulic fluid is released.
- The weight of the cage and the sliding ram forces the hydraulic fluid back into the reservoir, causing the ram to slide down and the cage to lower to the ground floor.

2. Suspended Hydraulic Lift



Working

- When hydraulic fluid under high pressure is admitted into the fixed cylinder of the jigger, the sliding ram is forced to move towards left.
- As one of the end of the sliding ram is connected to the movable pulley block.
- Hence the movable pulley block moves towards the left, thus increasing the distance between two pulley blocks.
- The wire rope connected to cage is pulled and the cage is lifted.
- For lowering the cage, water from fixed cylinder is taken out.
- The sliding ram moves towards right and hence movable pulley block also moves towards right.
- This decrease the distance between two pulley blocks and cage is lowered due to increased length of the rope

Thank
you